

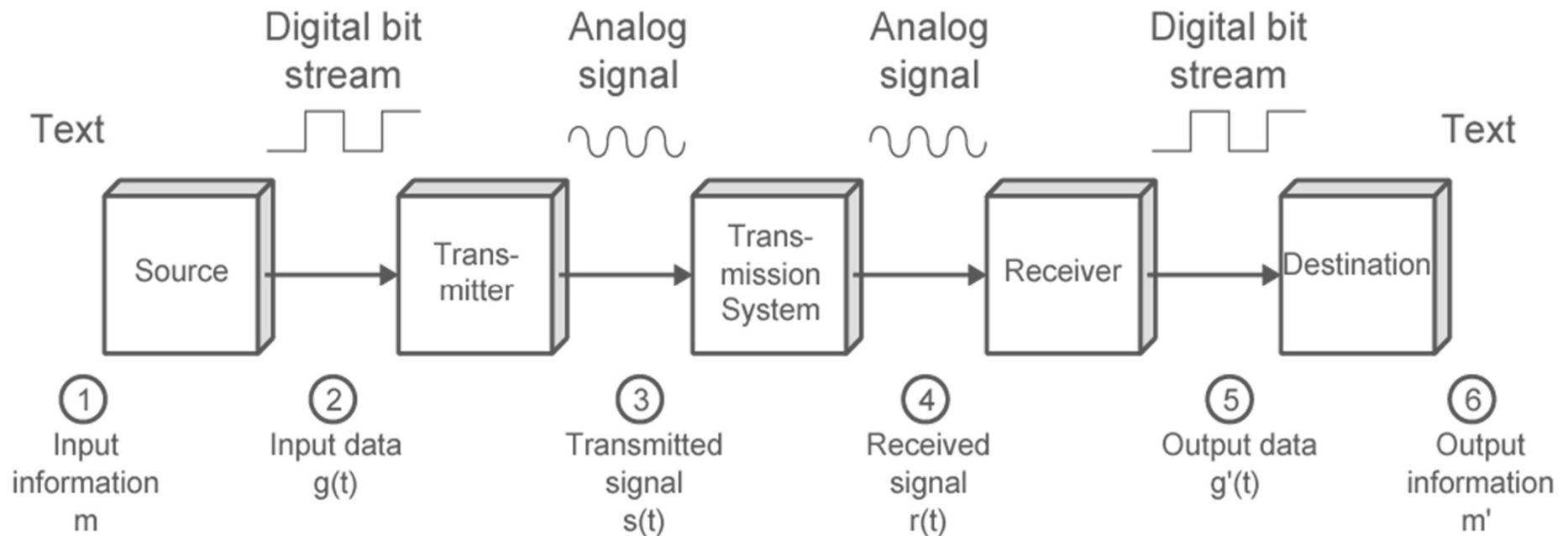
Data and Signals

Transforming Data into Signals

Physical

Data Communication Model

- Data must be transformed into electromagnetic signals in order to be transmitted.



Data vs. Signals

- Data: entities that convey meaning or information
 - Analog data: continuous information (e.g., human voice, temperature)
 - Digital data: discrete information (e.g., text, integers)
- Signal: electric or electromagnetic representation of data
 - Analog signal: continuously varying electromagnetic wave
 - Digital signal: sequence of discrete voltage pulses

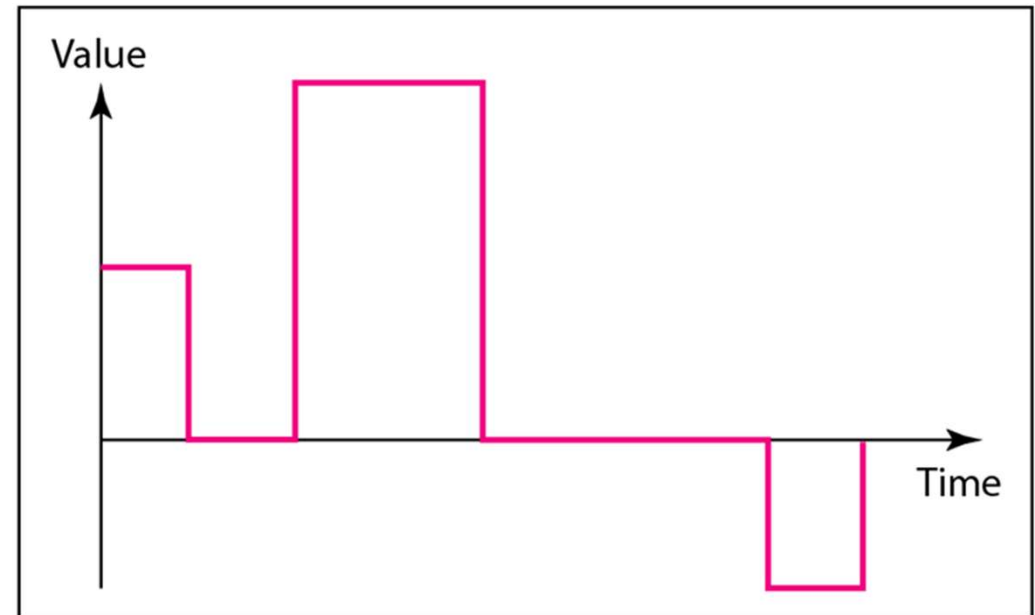
	Digital Signal	Analog Signal
Digital Data	Line coding (Ch. 2)	Modulation (Ch. 3)
Analog Data	PCM, digitization (Ch. 2)	amplitude/frequency modulation (Ch. 3)

Analog vs. Digital signals

- Analog signals: continuous
- Digital signals: discrete



a. Analog signal

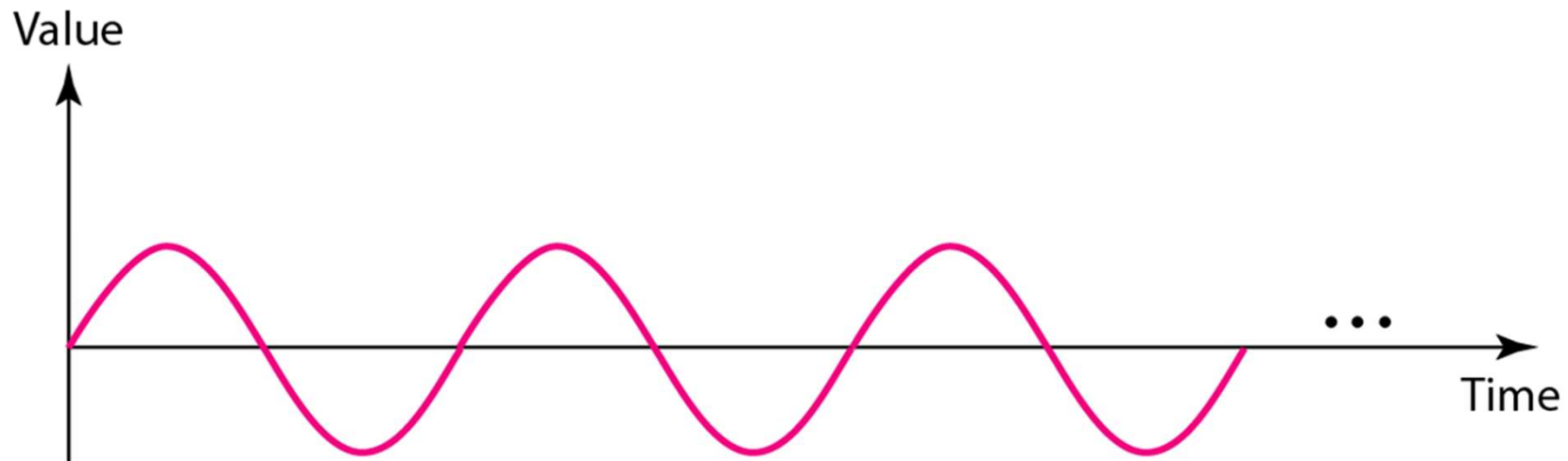


b. Digital signal

Analog Signals

Periodic Signals

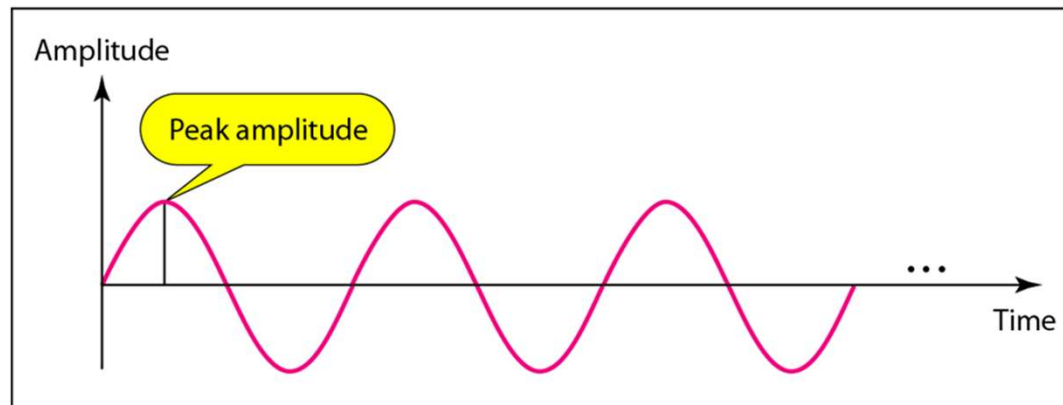
- The same pattern is repeated over time
- Sine wave



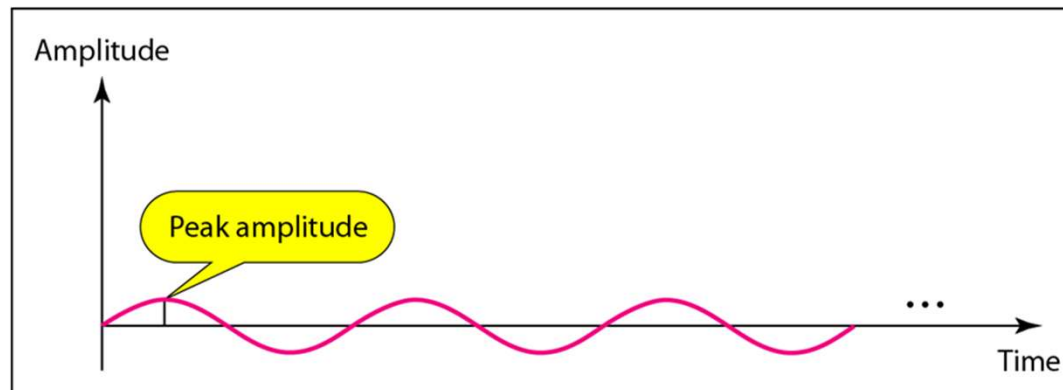
A simple sine wave

Characteristics of Periodic Signals

- Maximum amplitude
 - Absolute of the maximum value



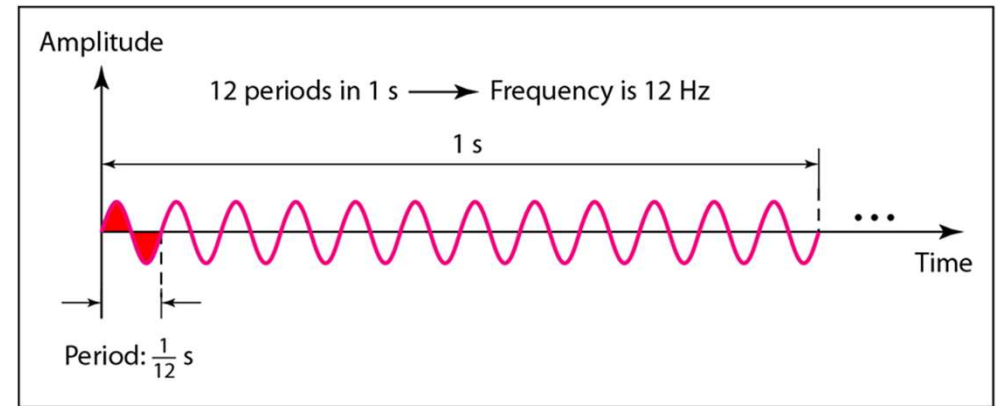
a. A signal with high peak amplitude



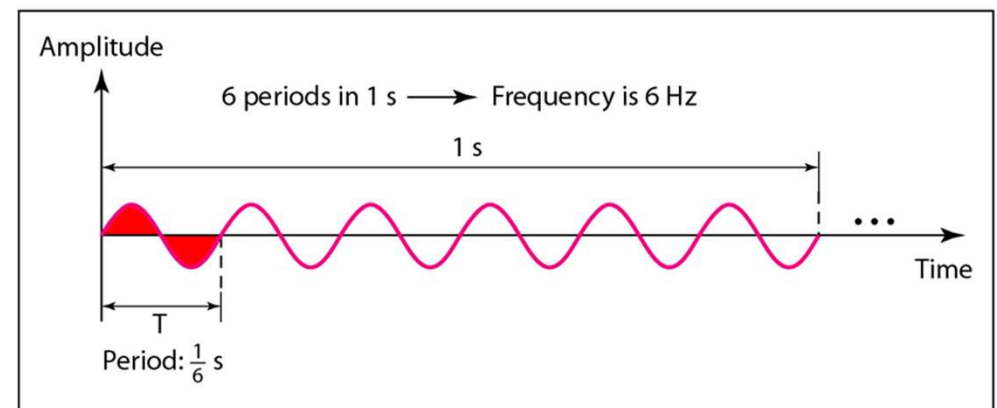
b. A signal with low peak amplitude

Characteristics of Periodic Signals

- Frequency (f)
 - How many times does the signal oscillate in a second?
 - Unit: Hz
- Period (T)
 - What is the time that the same pattern restarts?
 - Unit: s



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Characteristics of Periodic Signals

- Relation between frequency and period
 - frequency is the reciprocal of period

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Characteristics of Periodic Signals

- Units of period and frequency

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Exercise

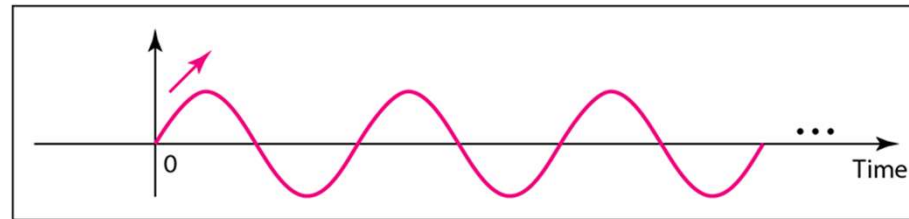
- Wi-Fi uses a frequency of 2.4GHz. What is the period of this signal?
- An FM radio station broadcasts at 93.1MHz. What is the period?
- The period of a signal is 100ms. What is its frequency in kilohertz?

Characteristics of Periodic Signals

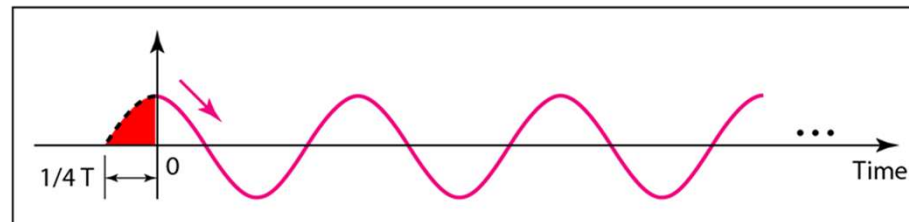
- Frequency indicates how fast a sine wave oscillates
 - Fast oscillation → high frequency
 - Slow oscillation → low frequency
- Extreme cases:
 - A signal that does not change over time
 - zero frequency, period is infinite
 - A signal that changes instantaneously
 - infinite frequency

Characteristics of Periodic Signals

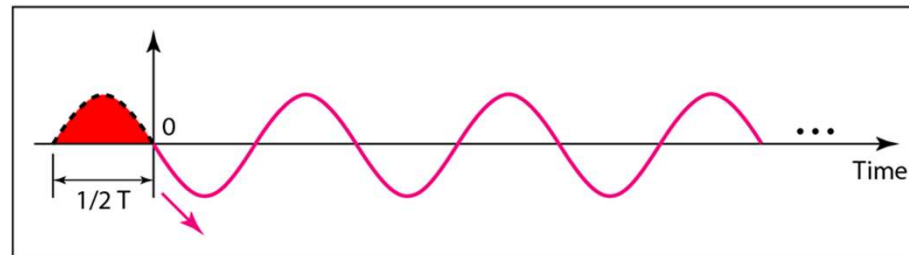
- Phase
 - The position of the waveform relative to time 0



a. 0 degrees



b. 90 degrees



c. 180 degrees

Characteristics of Periodic Signals

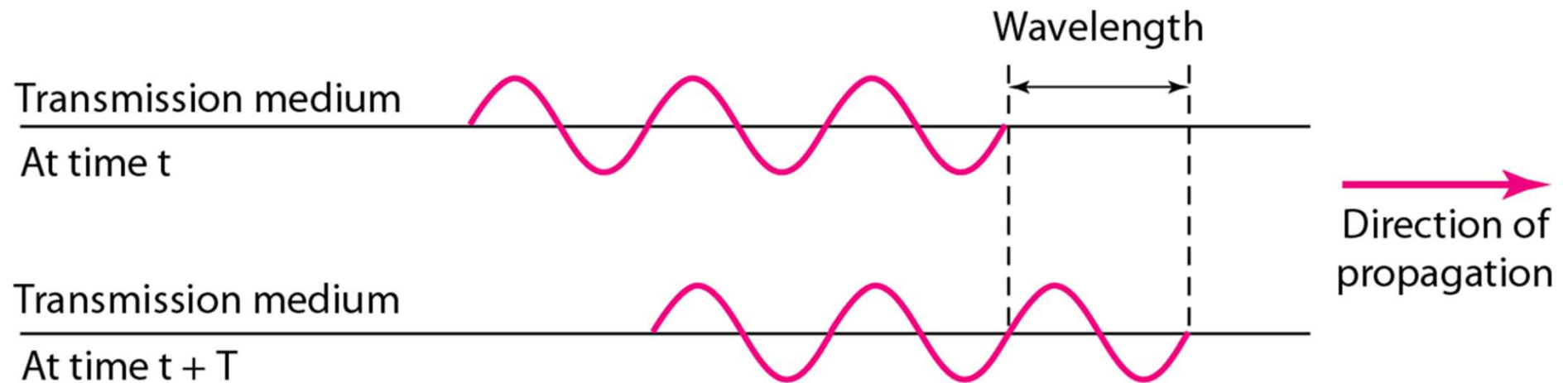
- Units of Phase: Degree vs. Radian
- Degree: $0 - 360^\circ$
- Radian: $0 - 2\pi$
- Example
 - $90^\circ = 1/2 \pi$
 - $180^\circ = \pi$
 - $270^\circ = 3/2 \pi$

Exercise

- A sine wave is offset $1/6$ cycle with respect to time 0. What is its phase in degrees and radians?
- Two sine waves have the same frequency and amplitude, but one is offset by $1/4$ cycle. What is the phase difference in degrees and radians?

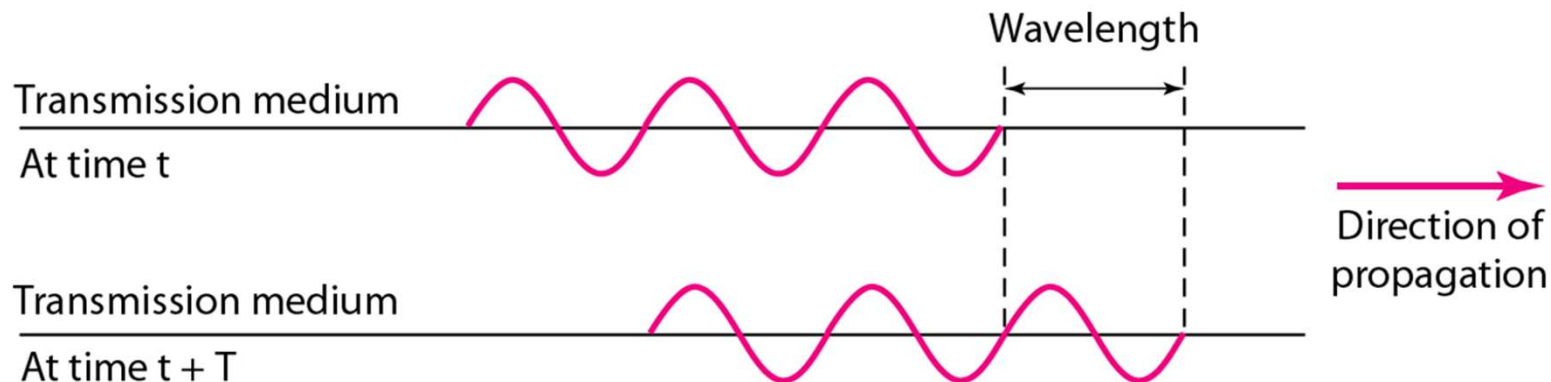
Characteristics of Periodic Signals

- wavelength (λ)
 - distance occupied by one cycle



Characteristics of Periodic Signals

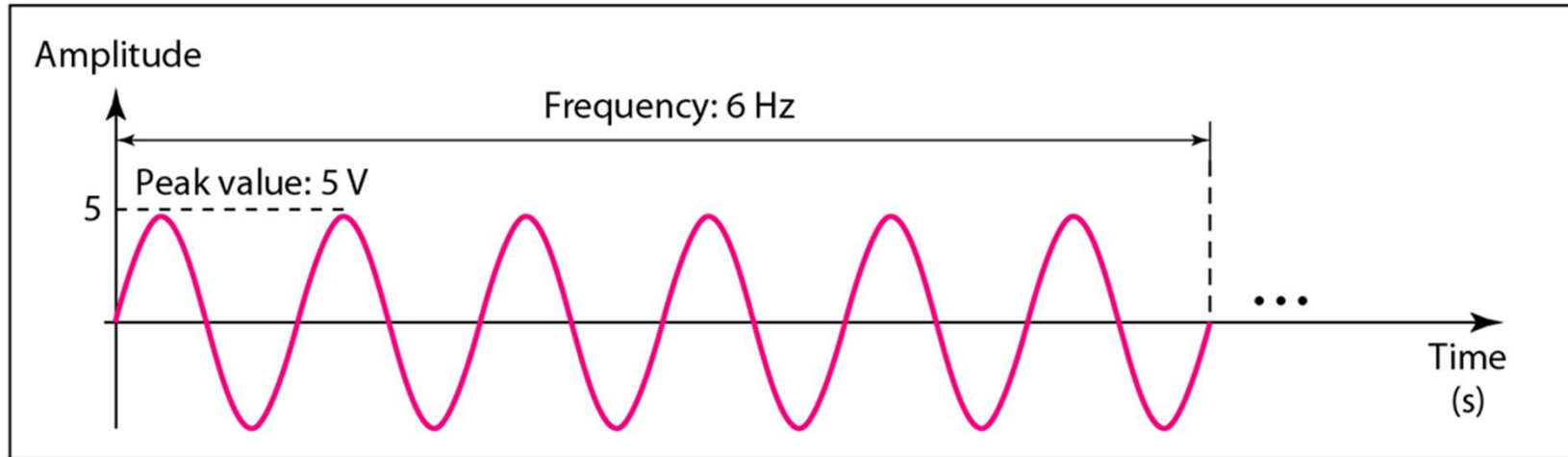
- wavelength (λ)
 - wavelength = propagation speed $\times$ period
= propagation speed / frequency
 - $\lambda = c / f$
 - c : propagation speed (depends on the medium)
 - $3 \times 10^8 \text{ m/s}$ in vacuum/air
 - $2 \times 10^8 \text{ m/s}$ in copper or fiber



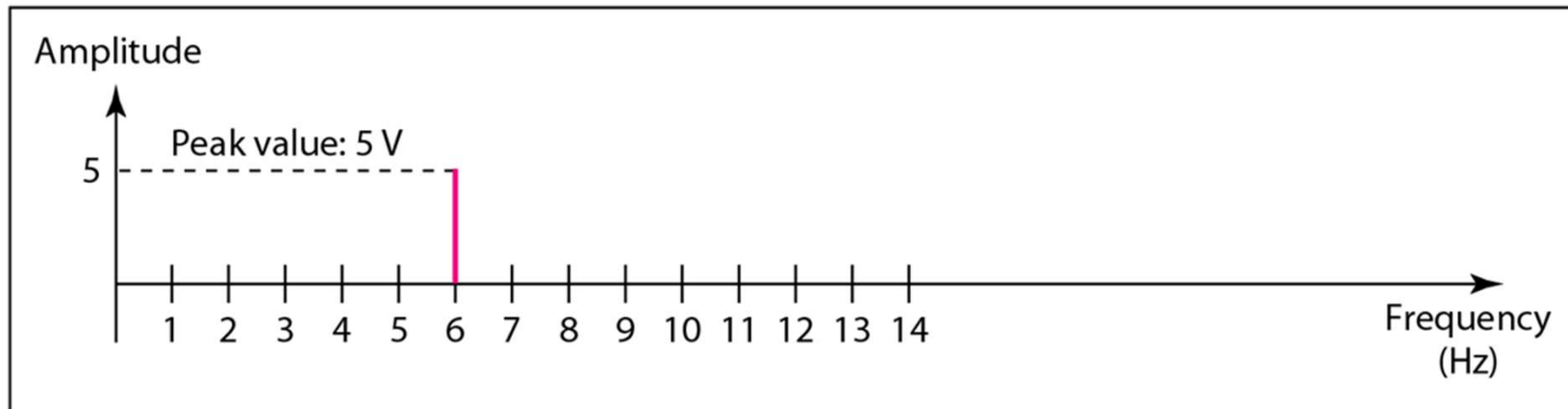
Time domain vs. Frequency domain

- Time-domain
 - The X-axis of a graph indicates time
- Frequency-domain
 - The X-axis of a graph indicates frequency

Time domain vs. Frequency domain



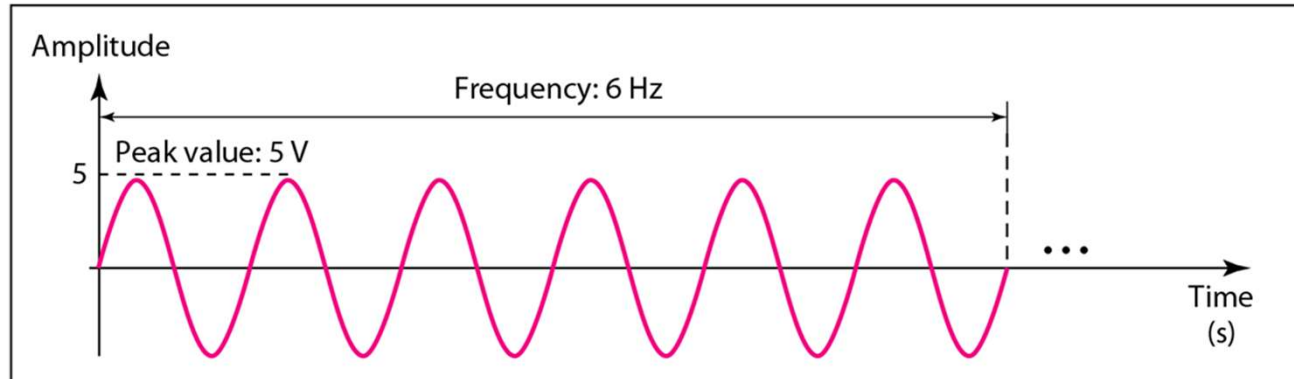
a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



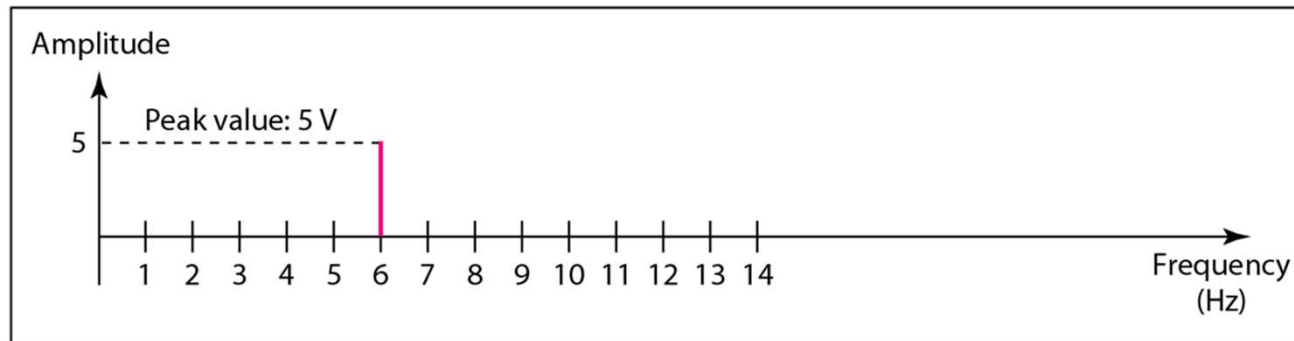
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Time domain vs. Frequency domain

- A simple sine wave becomes a single point in the frequency-domain graph



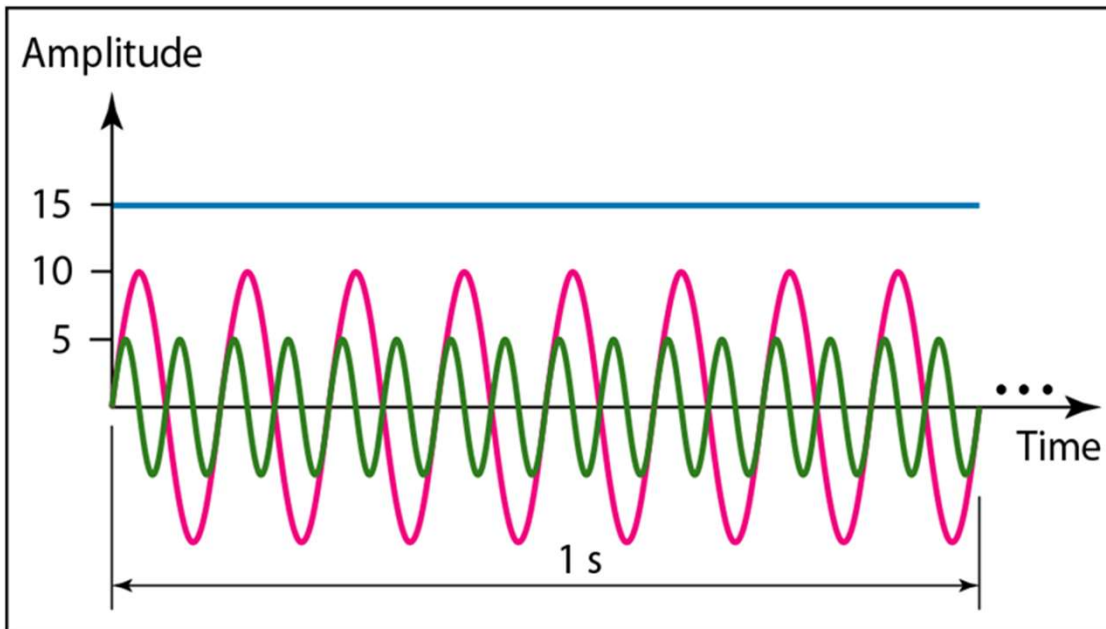
a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



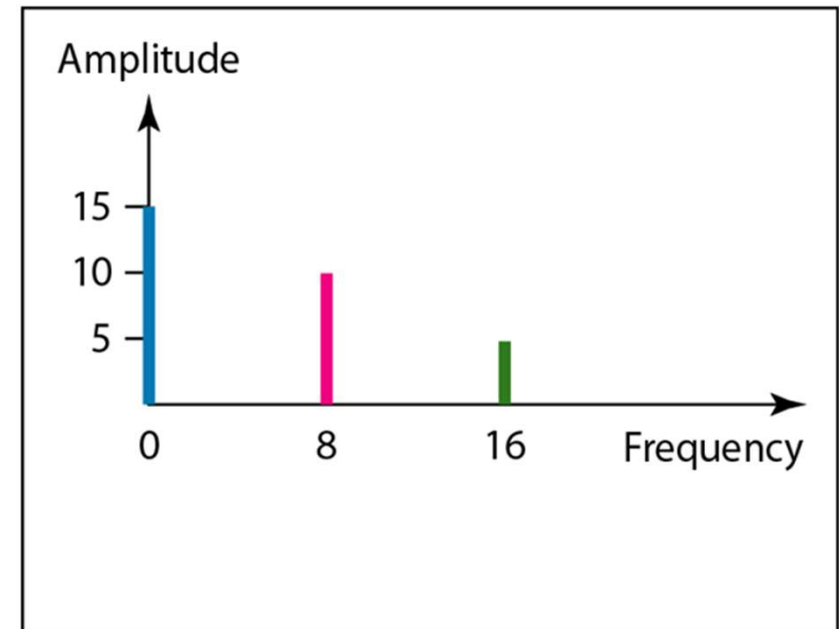
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Frequency domain: merit

- The frequency domain is more compact and useful when dealing with more than one sine wave



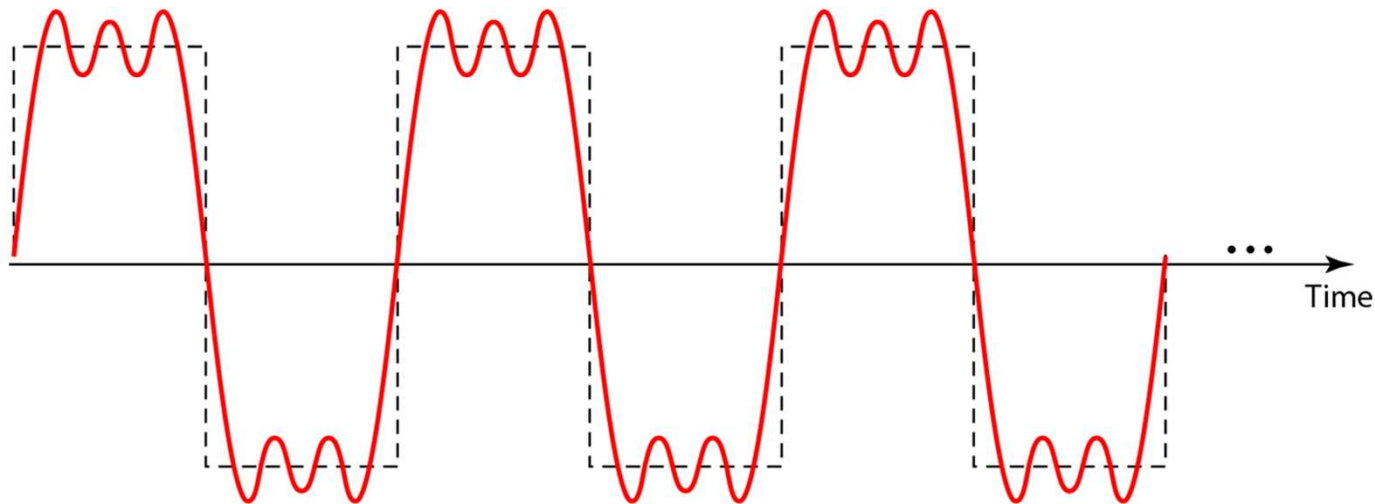
a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

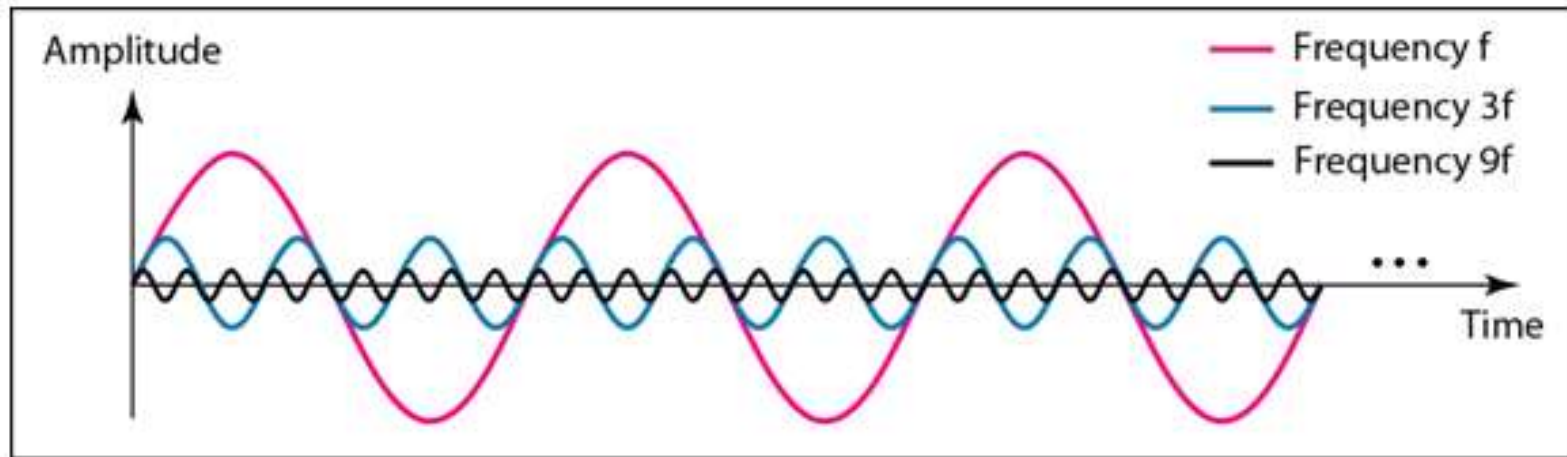
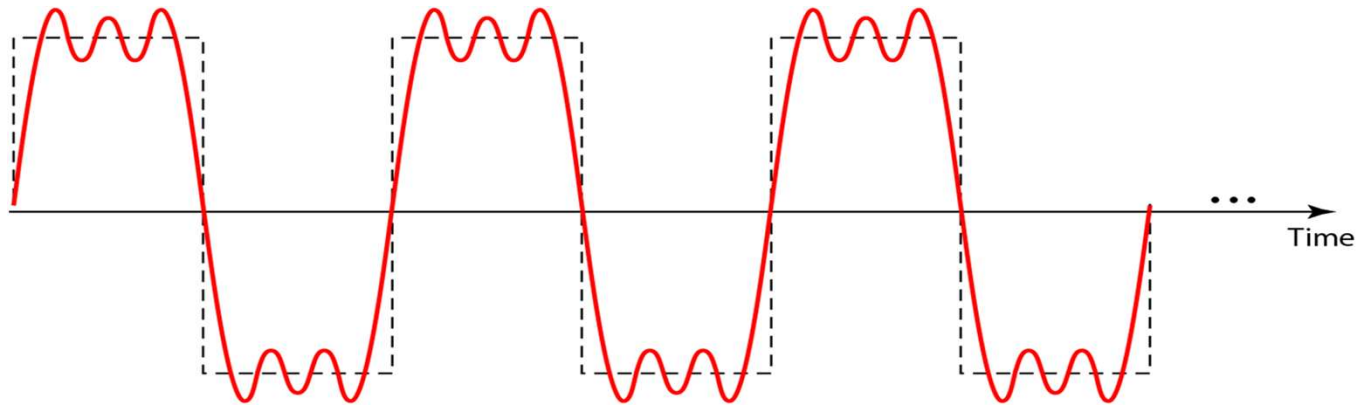
Composite Signals

- A composite periodic signal is a composition of multiple simple sine waves with different frequency, amplitude and phase
- Can be decomposed into simple sine waves



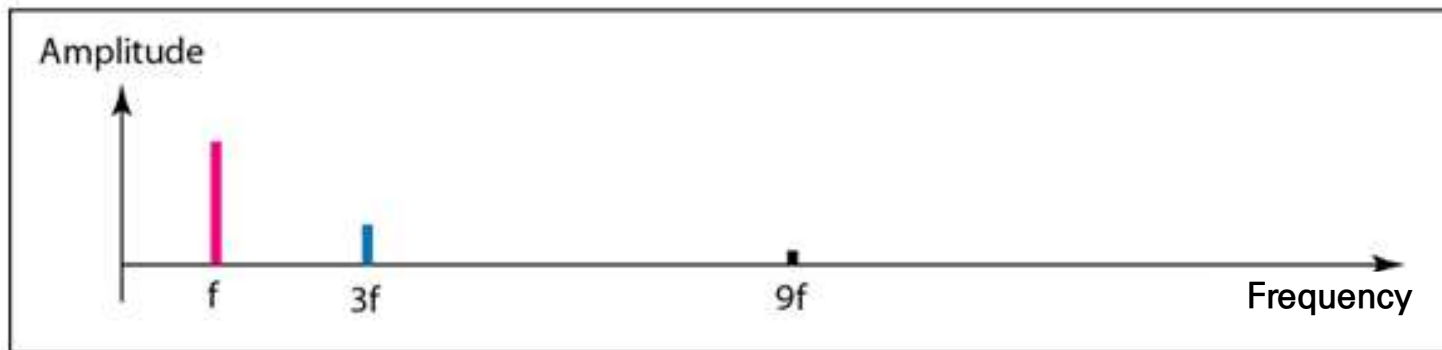
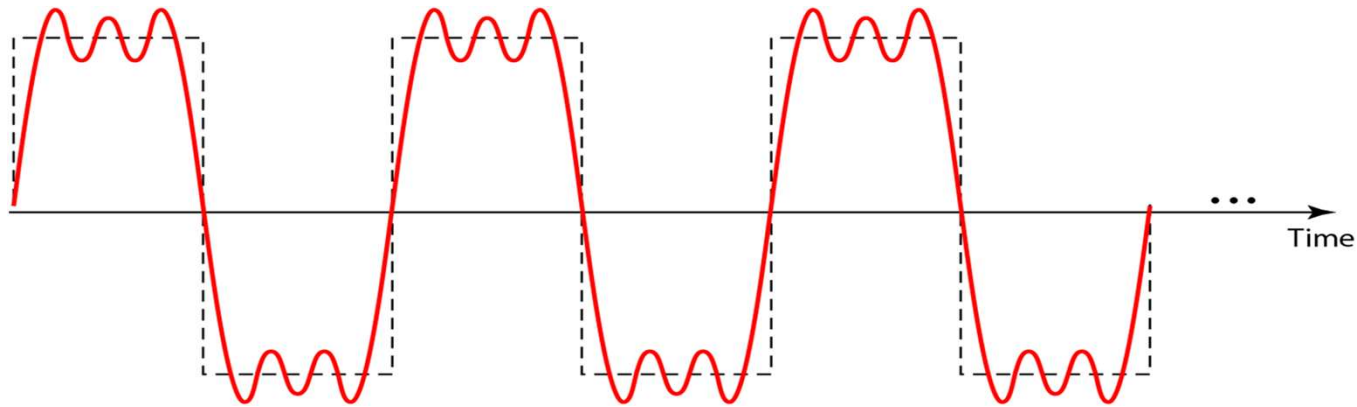
Composite Signals

- Time-domain decomposition



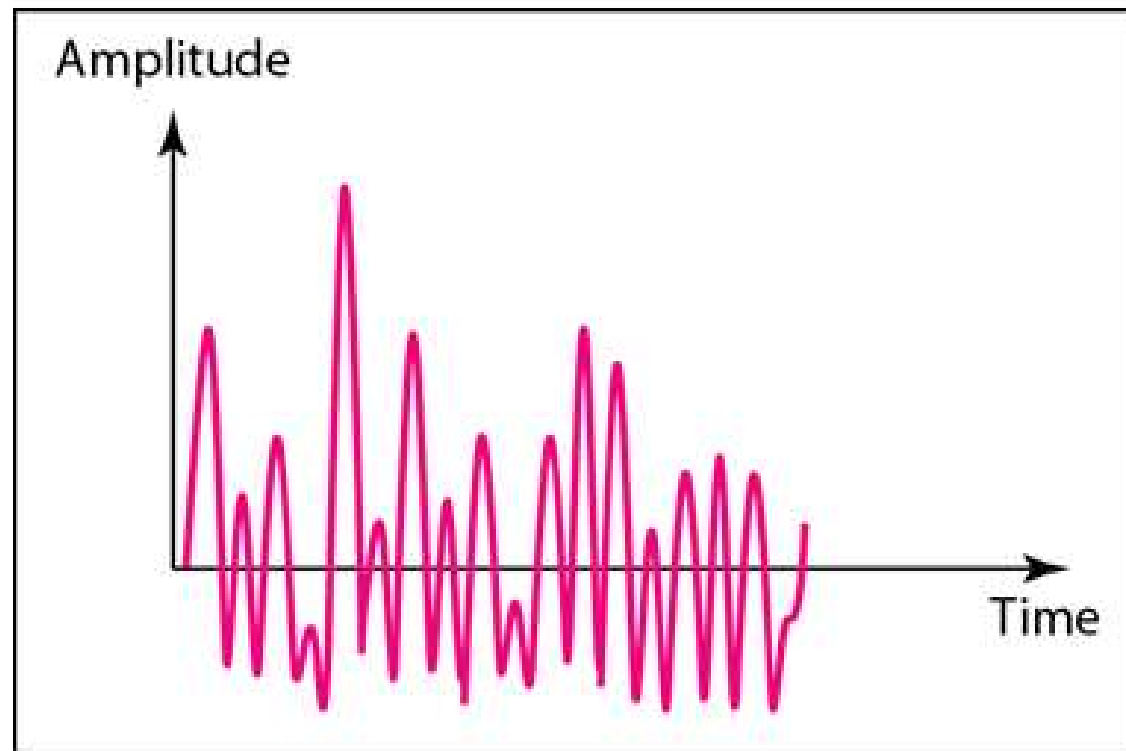
Composite Signals

- Frequency-domain decomposition



Composite Signals

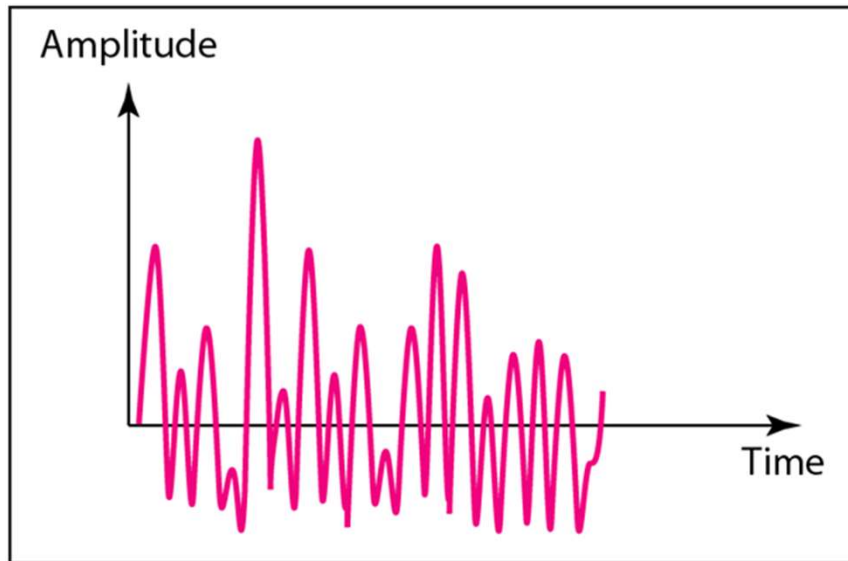
- aperiodic composite signal



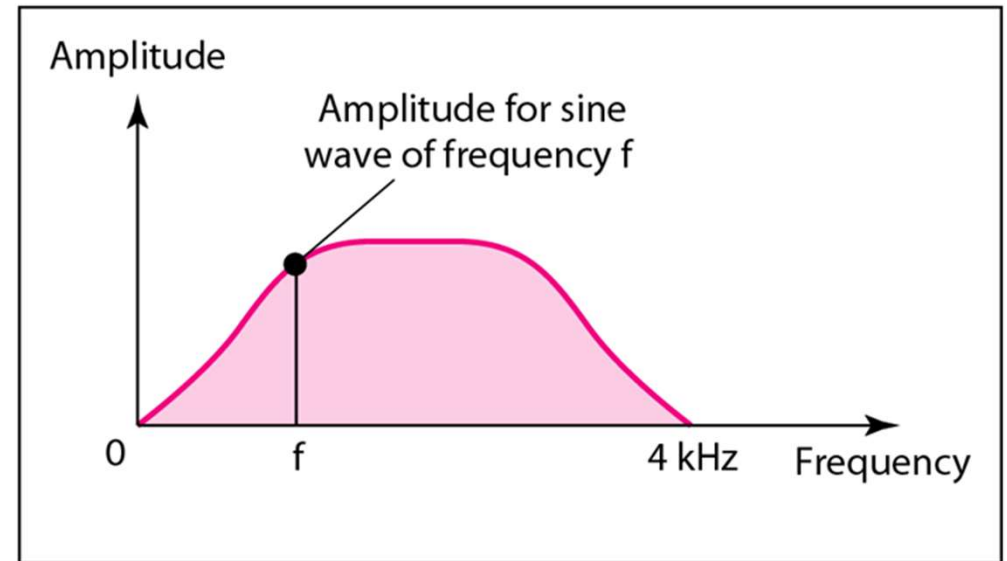
a. Time domain

Composite Signals

- frequency domain of aperiodic composite signal



a. Time domain



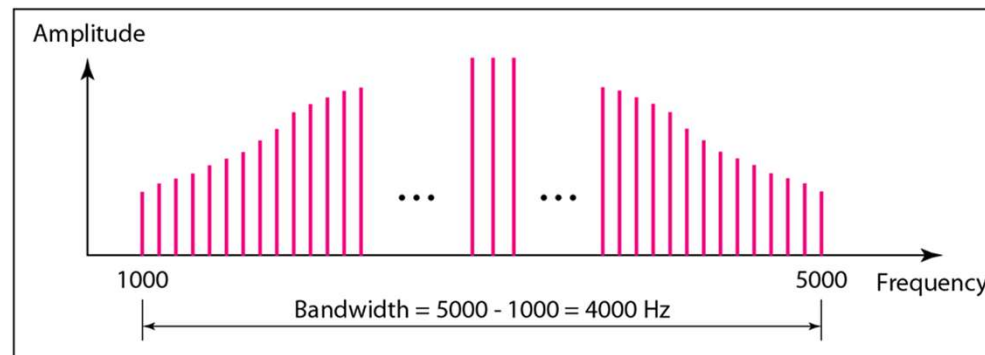
b. Frequency domain

Composite Signals

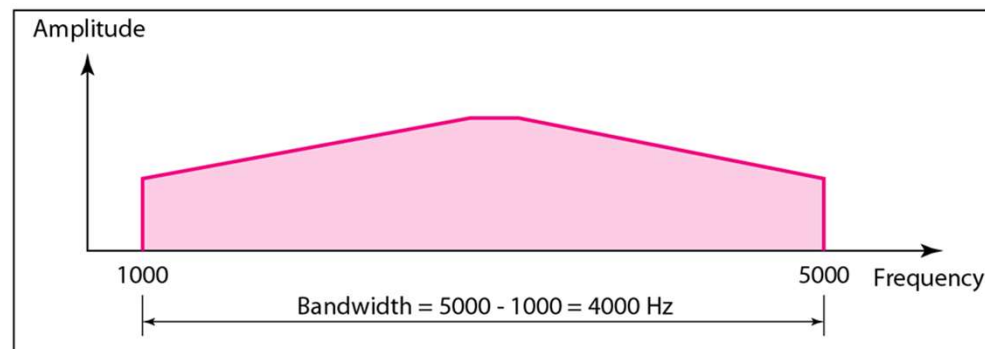
- If the composite signal is periodic, it will appear in frequency domain as composition of signals with discrete frequencies
- If the composite signal is aperiodic, it will appear in frequency domain as composition of signals with continuous frequencies

Bandwidth

- Bandwidth is the range of frequencies contained in a composite signal
 - difference between the highest frequency and the lowest frequency



a. Bandwidth of a periodic signal



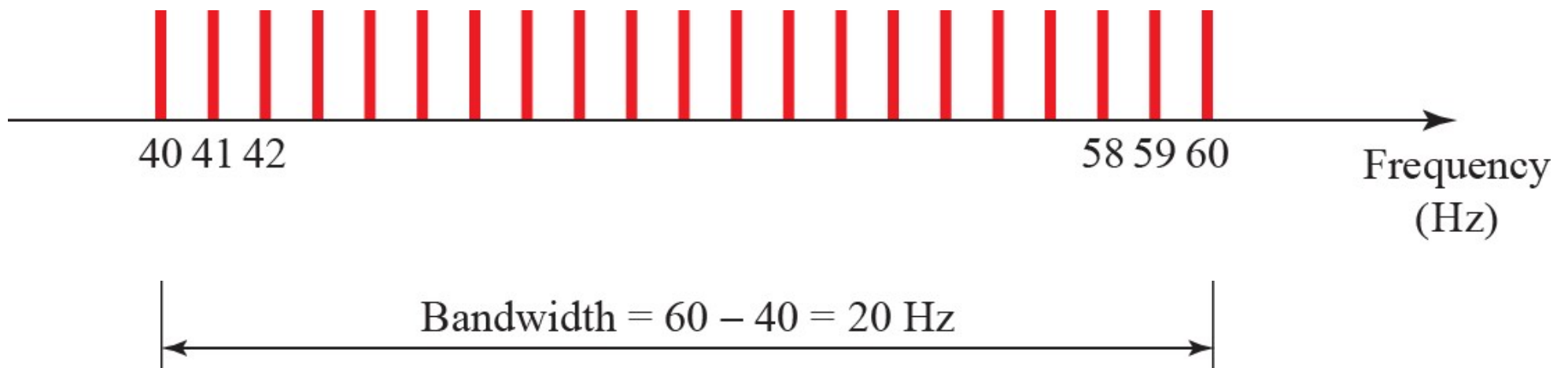
b. Bandwidth of a nonperiodic signal

Exercise

- If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900Hz, what is its bandwidth?
- Draw the spectrum, assuming all components have a maximum amplitude of 10V.

Exercise

- A periodic signal has a bandwidth of 20Hz. The highest frequency is 60Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.



Aperiodic composite signals

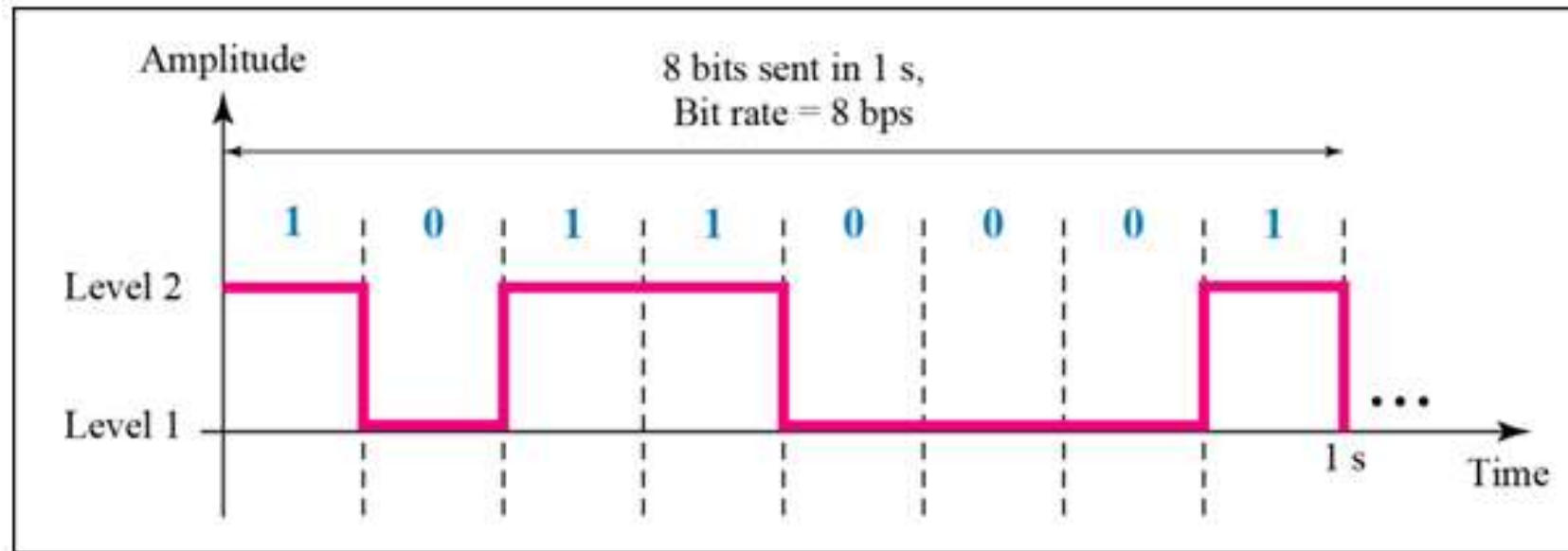
- AM radio
 - Each channel has 10kHz bandwidth
 - 530 – 1700kHz band
- FM radio
 - Each channel has 200kHz bandwidth
 - 88 – 108MHz band
- Wi-Fi 802.11n
 - 20MHz or 40MHz bandwidth in 2.4GHz band
- LTE
 - 10MHz or 20MHz bandwidth in 1.8GHz band

Digital Signals

Digital Signals

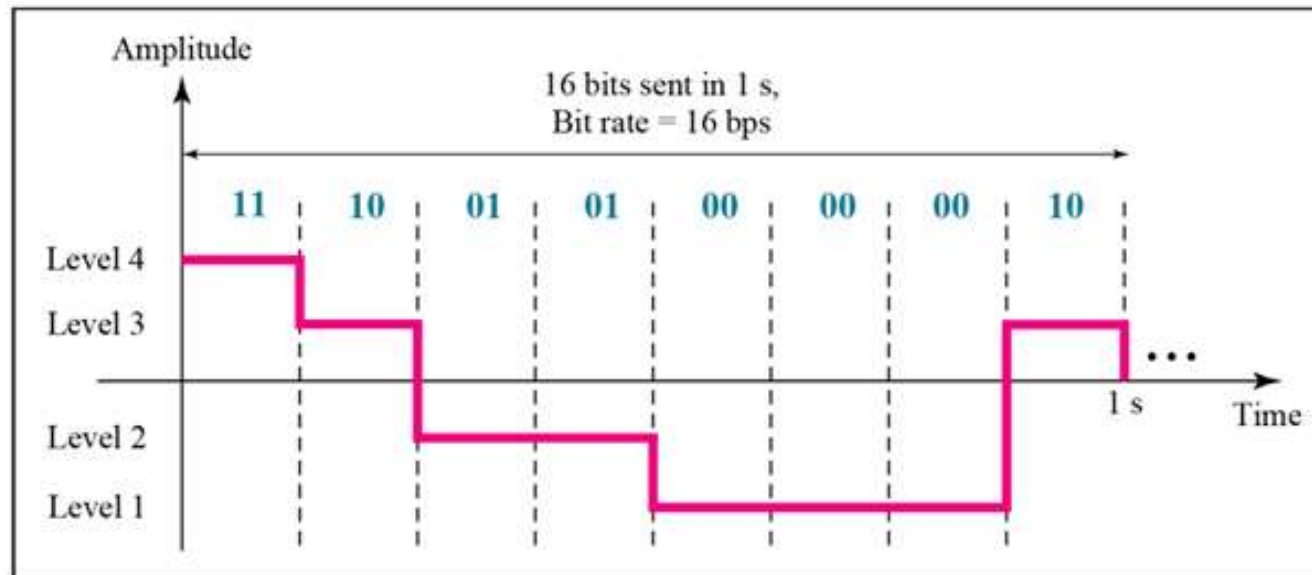
- Signals that contain bit information (0, 1) using amplitude, frequency and phase
- Example
 - Bit 1 $\rightarrow$ 5V
 - Bit 0 $\rightarrow$ 0V

Digital Signals



Digital Signals

- Signal levels can be more than two
 - 4 levels can contain two bits of information
 - Level 1: 00
 - Level 2: 01
 - Level 3: 10
 - Level 4: 11



Exercise

- A digital signal has eight levels. How many bits can each level represent?
- We want to transmit 6 bits per level change. How many signal levels are needed?

Transfer rate (bit rate)

- How fast the data is transferred
- Unit: bps (bits per second)
- 1 Kbps = 1000 bps
- 1 Mbps = 1000 Kbps
- 1 Gbps = 1000 Mbps

Exercise

- Assume we need to download text documents at the rate of 100 pages per second. Each page contains 24 lines, and each line contains 80 characters. Each character uses 1 byte. What is the required bit rate of the channel?

Exercise

- What is the bit rate for high-definition TV (HDTV), which has the following specification?
 - 1920 x 1080 pixels per screen
 - 30 frames per second
 - 1 pixel uses 24 bits (RGB)

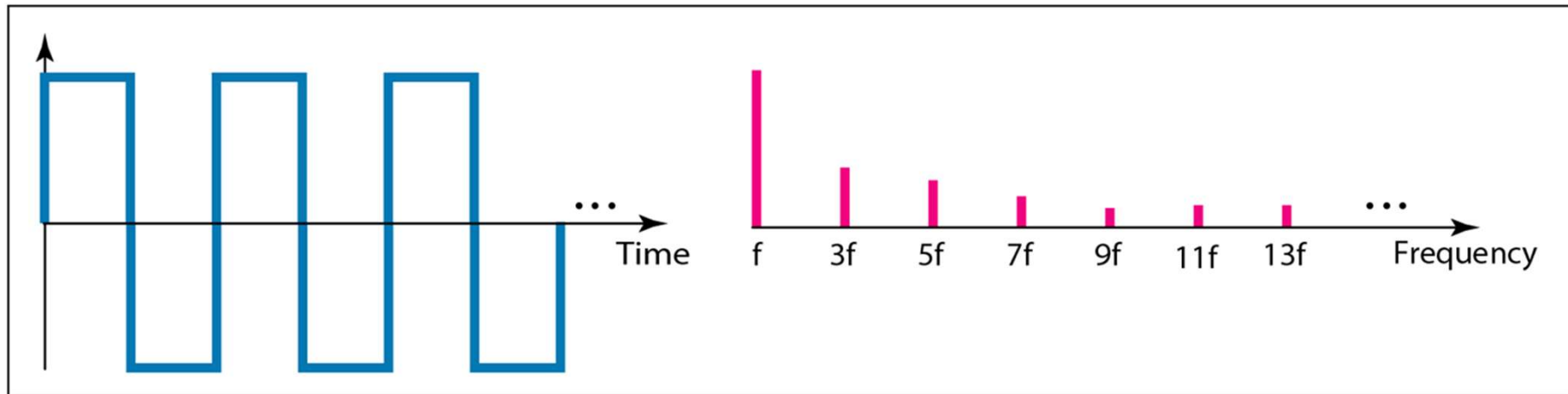
Transmission of Digital Signals

Digital Signals

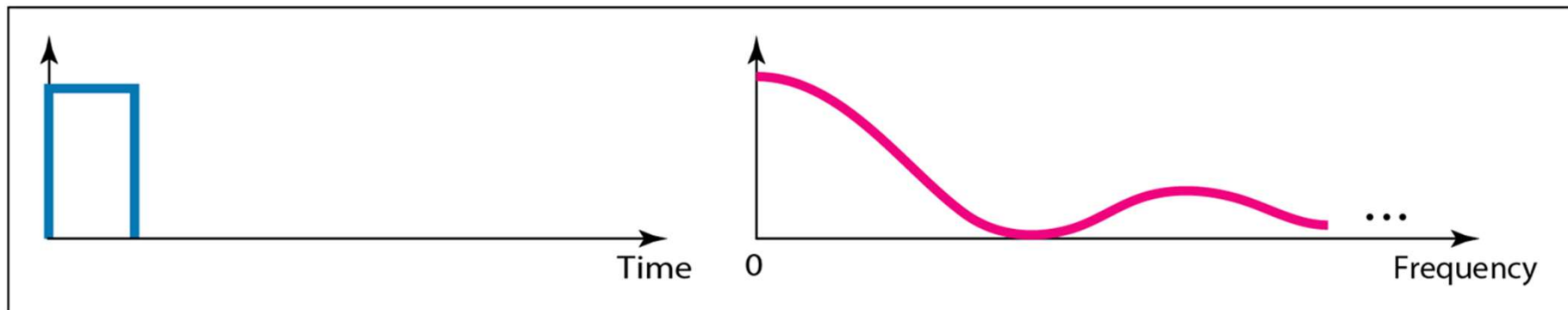
- How do we transmit digital signals over media?

Digital Signals

- Time domain and frequency domain of a digital signal



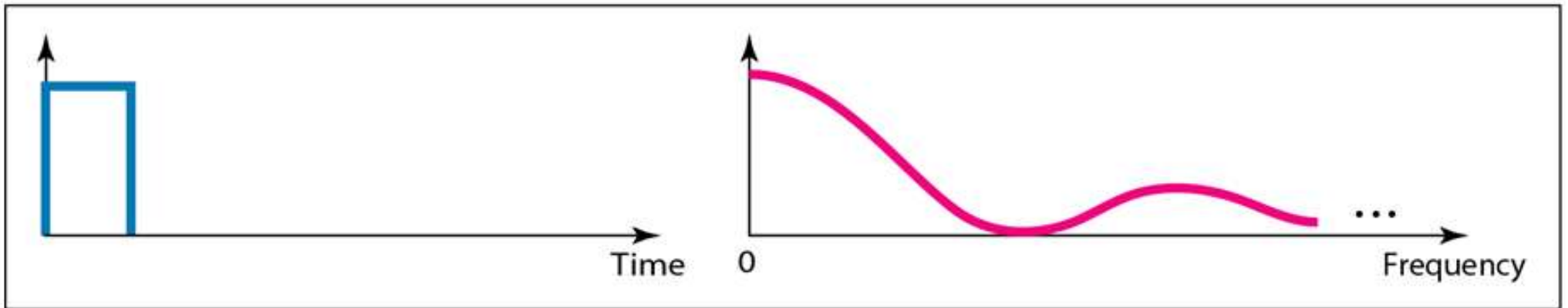
a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

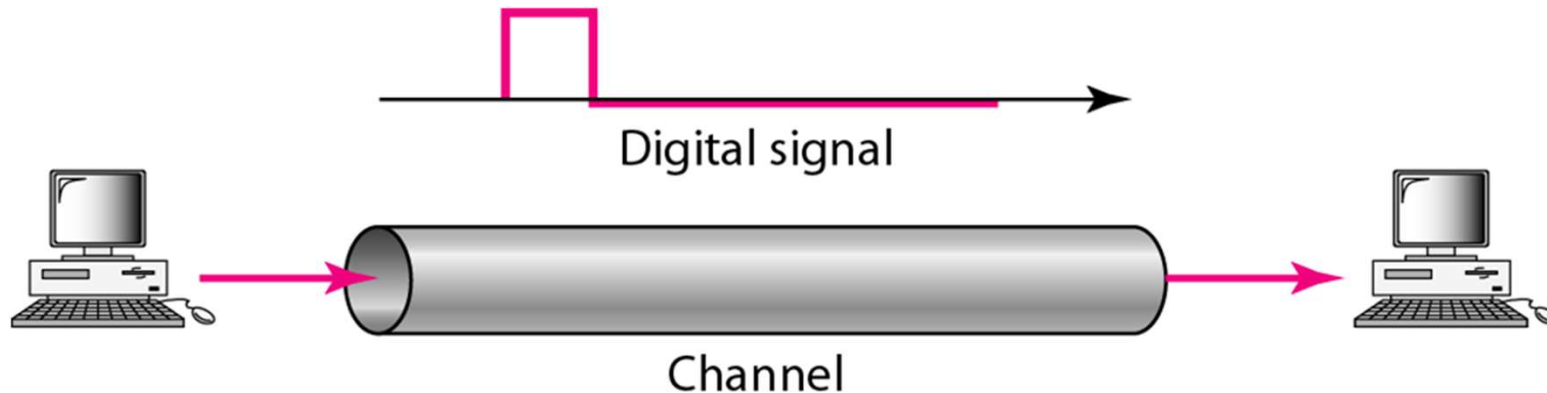
Digital Signals

- A digital signal is a composite analog signal with infinite bandwidth



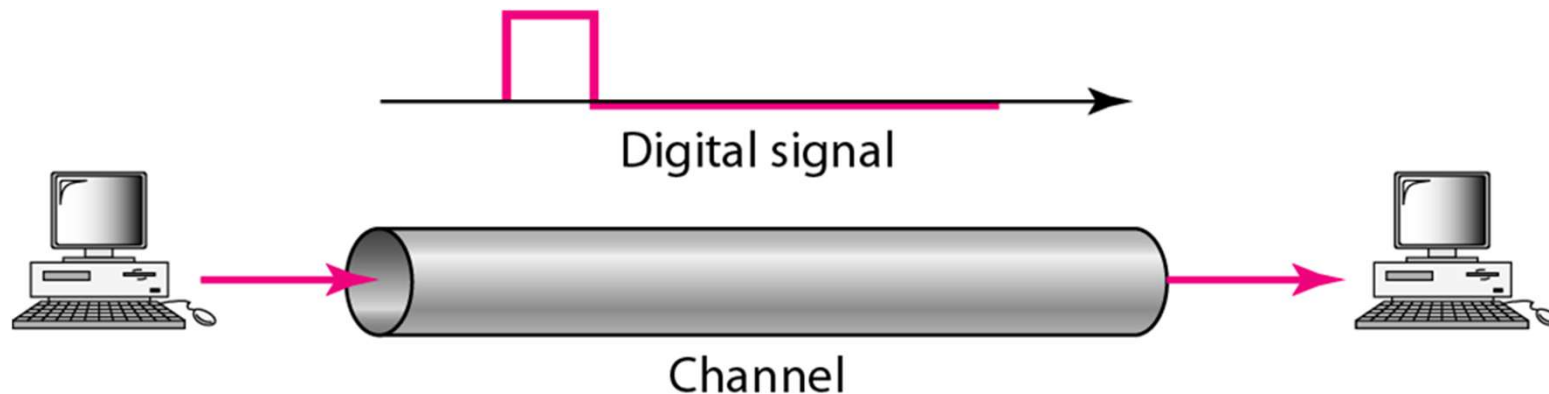
Baseband Transmission

- The simplest way to transmit a digital signal is to send it directly over the channel
- This is called baseband transmission
 - The signal occupies frequencies starting from 0



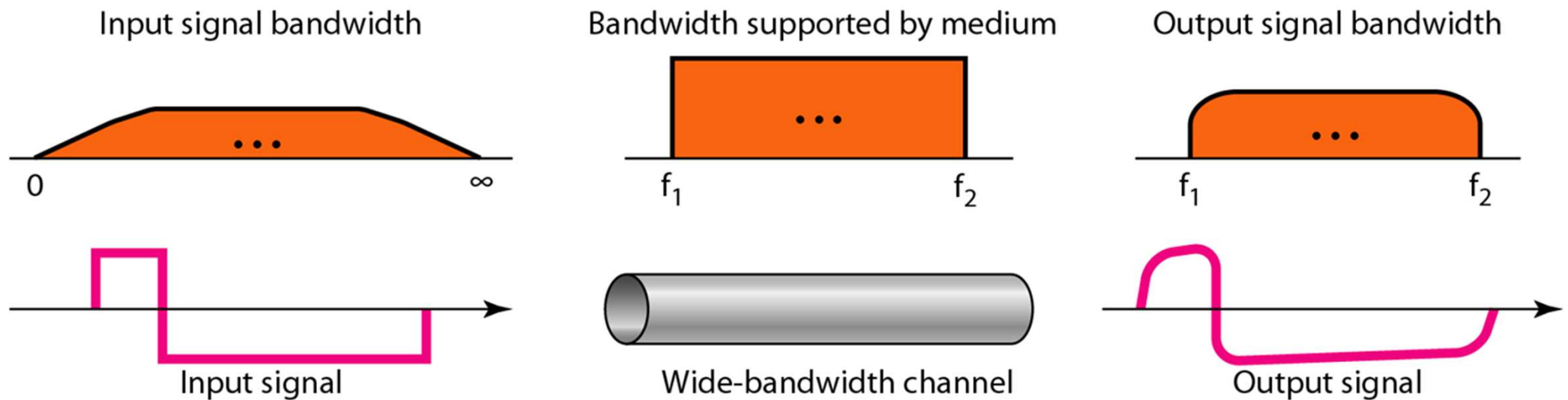
Baseband Transmission

- A digital signal has frequency components starting from 0 (DC) to infinity
- Baseband transmission requires a low-pass channel (a channel that passes frequencies from 0)
- Ideally, this channel would need infinite bandwidth to transmit the signal perfectly



Baseband Transmission

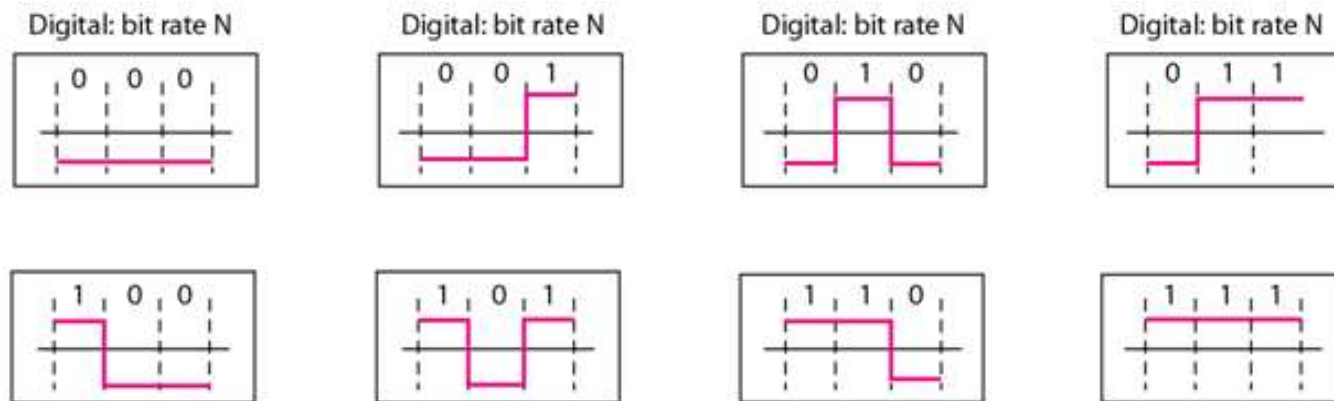
- Real-life media cannot have infinite bandwidth
- Some frequency components are cut from the original signal, leaving an approximate signal
- The more bandwidth we use, we get an approximate signal that is closer to the original



For baseband transmission, $f_1 = 0$ (low-pass channel)

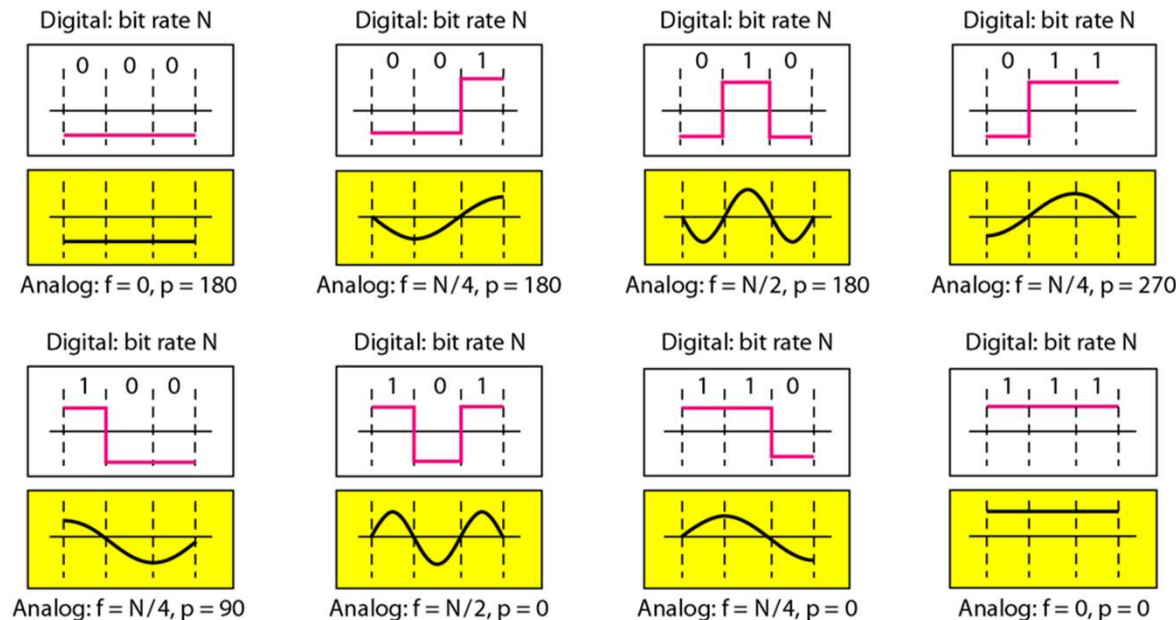
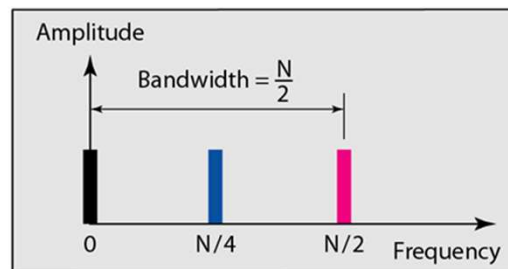
Approximation of Digital Signals

- Suppose we encode 3 bits into the following digital signals (8 possible patterns)
- Transmitting these signals perfectly requires infinite bandwidth
- However, if we can still distinguish all 8 patterns with limited bandwidth, the receiver can recover the original bits



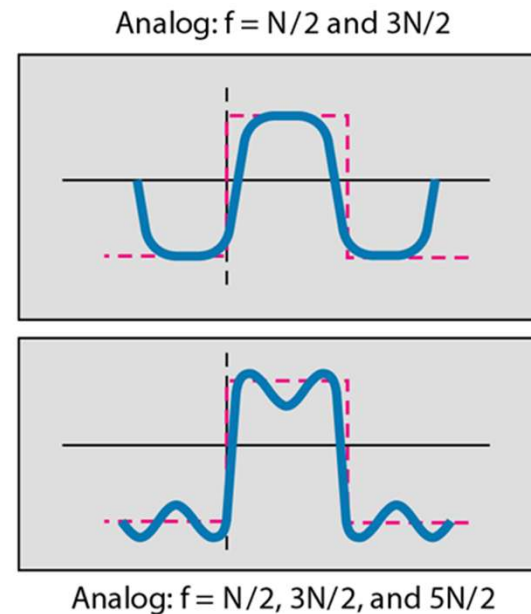
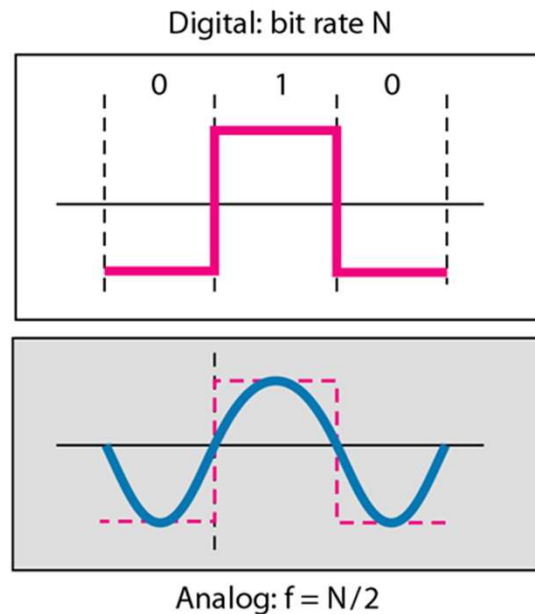
Approximation of Digital Signals

- We can approximate these signals using limited bandwidth
- For 2-level signals, the minimum bandwidth required is $N/2$ Hz, where N is the bit rate



Approximation of Digital Signals

- If we use more bandwidth, we can create better approximates



Approximation of Digital Signals

- Basically, bandwidth needed for baseband transmission is proportional to transfer rate
- More bandwidth provides better approximation – increased accuracy

<i>Bit Rate</i>	<i>Harmonic 1</i>	<i>Harmonics 1, 3</i>	<i>Harmonics 1, 3, 5</i>
$n = 1 \text{ kbps}$	$B = 500 \text{ Hz}$	$B = 1.5 \text{ kHz}$	$B = 2.5 \text{ kHz}$
$n = 10 \text{ kbps}$	$B = 5 \text{ kHz}$	$B = 15 \text{ kHz}$	$B = 25 \text{ kHz}$
$n = 100 \text{ kbps}$	$B = 50 \text{ kHz}$	$B = 150 \text{ kHz}$	$B = 250 \text{ kHz}$

Summary: Baseband Transmission

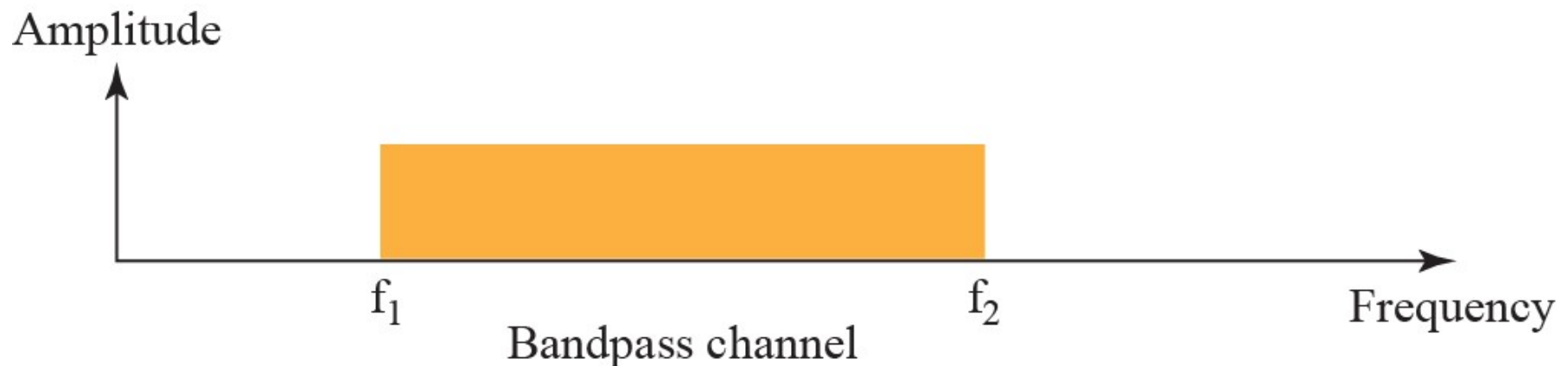
- Baseband transmission: transmitting a digital signal using frequencies starting from 0 (DC)
- To transmit undistorted digital signal, we need infinite bandwidth (which is impossible)
- Transfer rate is proportional to bandwidth
- For 2-level signals, to send data at N bps, we need at least $N/2$ bandwidth
 - If we have a bandwidth f , maximum transfer rate is $2f$

Exercise

- What is the required bandwidth of a low-pass channel if we need to send 1Mbps by using baseband transmission with 2 signal levels?
- We have a low-pass channel with bandwidth 100kHz. What is the maximum bit rate of this channel using 2 signal levels?

Bandpass channel

- Passes frequencies from f_1 to f_2 ($f_1 > 0$)
- Cannot pass low frequencies (near 0), unlike low-pass channels
- Used in wireless communications
 - signals are modulated onto a carrier frequency
 - low-pass channel: twisted pair, coaxial cable
- Bandwidth = $f_2 - f_1$



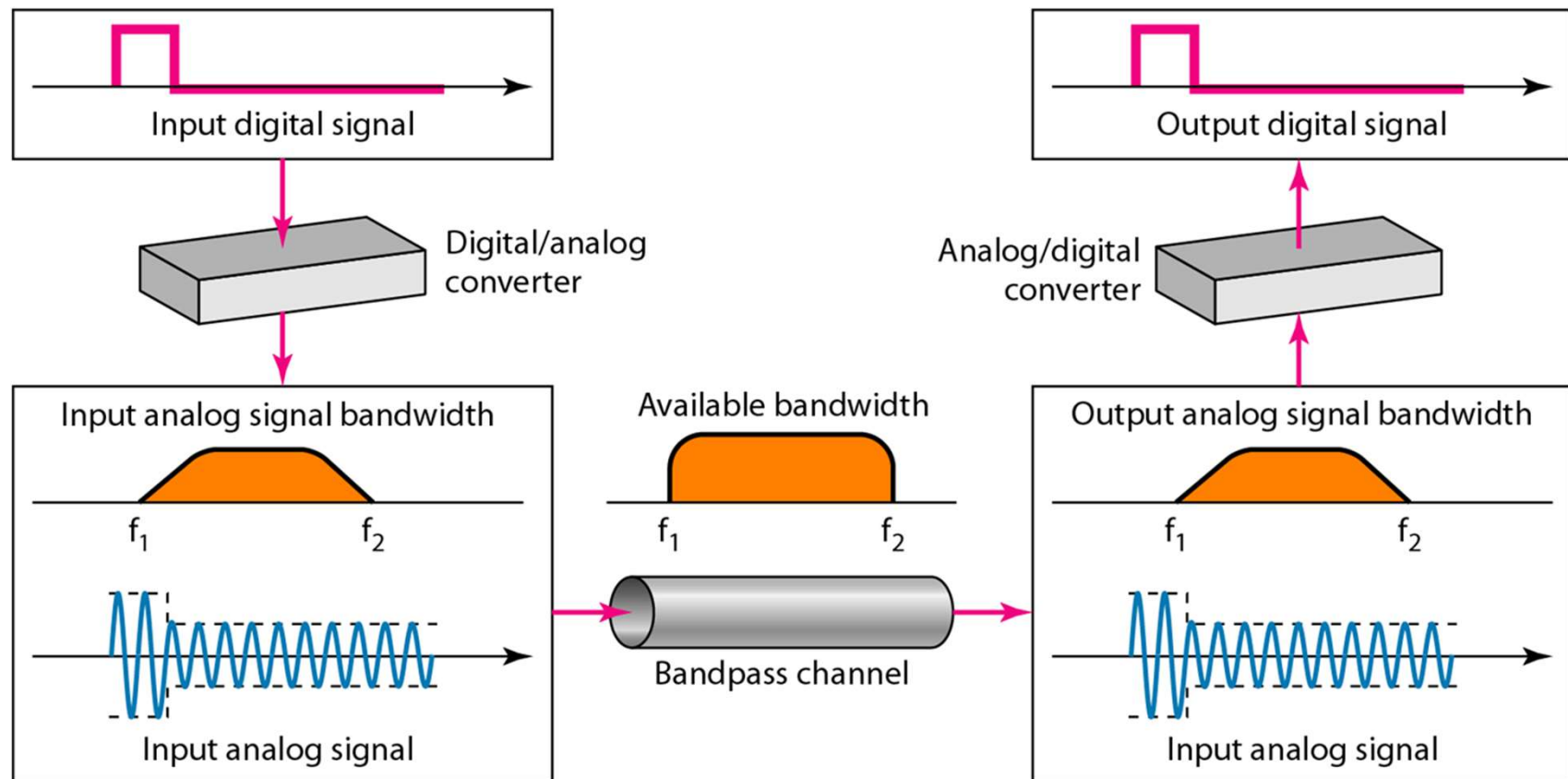
Example: mobile communication in Korea

- SKT
 - 800MHz: LTE (20MHz)
 - 1.8GHz: LTE (35MHz)
 - 2.1GHz: LTE (30MHz) + 3G (10MHz)
 - 2.6GHz: LTE (60MHz)
 - 3.6GHz: 5G NR (100MHz)
- KT
 - 900MHz: LTE (20MHz)
 - 1.8GHz: LTE (20MHz)
 - 2.1GHz: LTE (40MHz)
 - 3.5GHz: 5G NR (100MHz)
- LG U+
 - 800MHz: LTE (20MHz)
 - 1.8GHz: LTE (20MHz)
 - 2.1GHz: LTE (20MHz)
 - 2.6GHz: LTE (40MHz)
 - 3.4GHz: 5G NR (100MHz)



Bandpass Transmission using Modulation

- To transmit signals through a bandpass channel, we need to shift the signal to the channel's frequency range using modulation



Baseband vs. Bandpass in Practice

- Broadband (low-pass channel)
 - Ethernet over twisted pair (e.g., 100BASE-TX, 1000BASE-T)



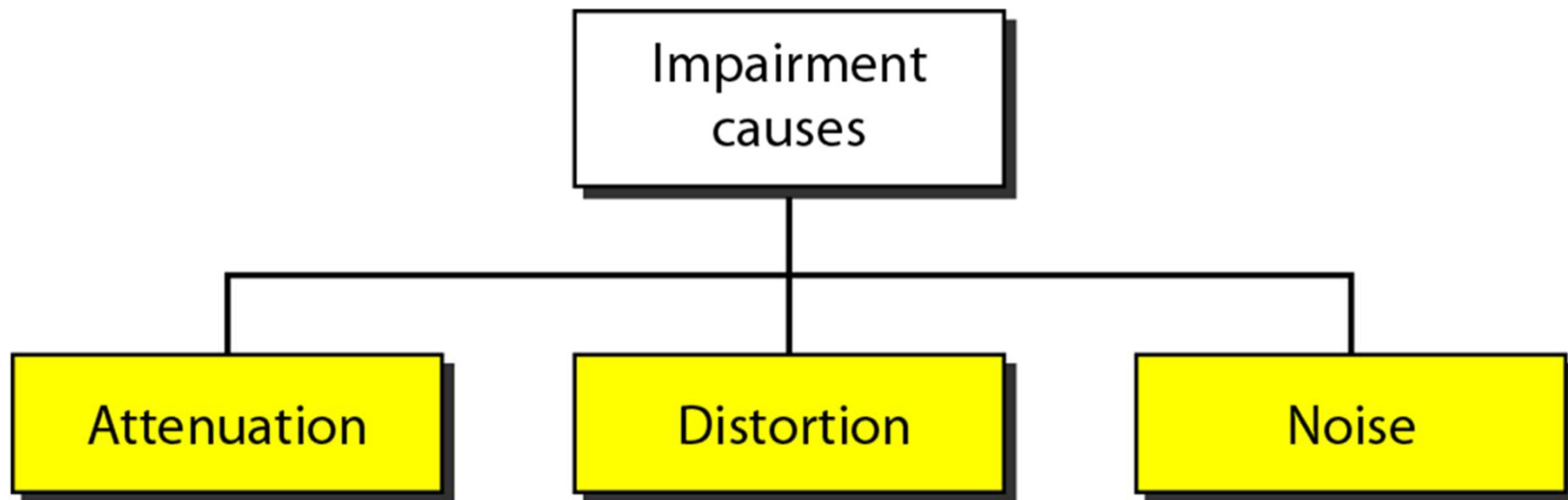
- Bandpass
 - Wi-Fi (2.4/5GHz wireless channel)
 - LTE/5G (cellular wireless channel)



Transmission Impairment

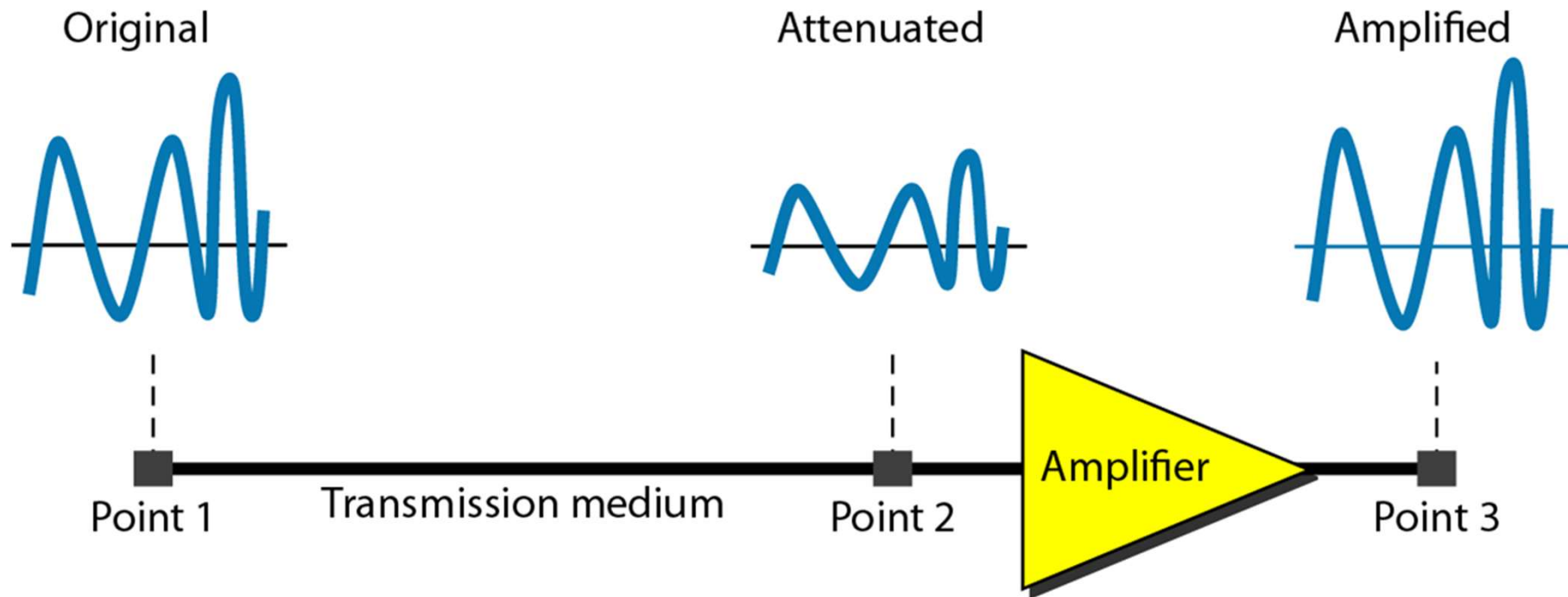
Transmission Impairment

- Imperfect media causes signal impairment
- The signal at the beginning of the medium is not the same as the signal at the end of the medium



Attenuation

- Loss of energy (signal power) as the signal travels through media
- Analog signals: amplifiers can boost the signal, but also amplify noise
- Digital signals: repeaters can regenerate the original signal, eliminating accumulated noise



Units for signal strength: decibel (dB)

- $\text{dB} = 10 \log_{10}(P_2/P_1)$
 - Indicates the ratio of two signal strength
 - Positive dB $\rightarrow$ signal gained (amplified)
 - Negative dB $\rightarrow$ signal lost (attenuated)
- Why use dB?
 - Signal power can vary by factors of millions
 - Log scale makes large ranges manageable
 - dB values can be simply added along a transmission path

Example

- Suppose a signal travels through a transmission media and its power is reduced to one half. This means that $P_2 = 0.5P_1$. In this case, the attenuation can be calculated as:

$$10 \log_{10} P_2/P_1 = 10 \log_{10} (0.5 P_1) / P_1 = 10 \log_{10} 0.5 = 10 \times (-0.3) = -3 \text{ dB.}$$

- A loss of 3dB (-3dB) is equivalent to losing one-half the power

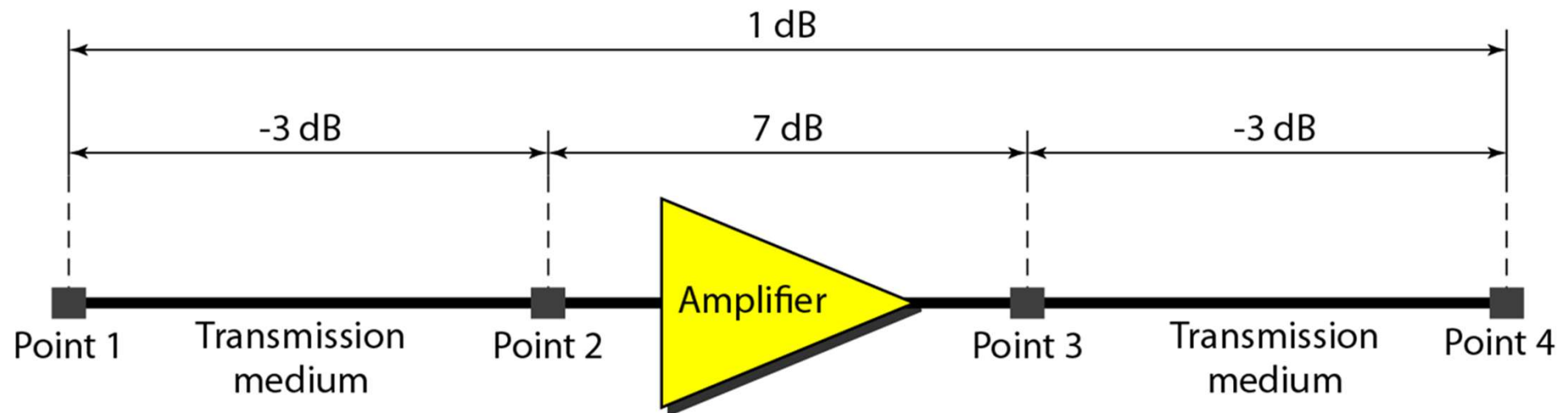
Example

- A signal travels through an amplifier and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification can be calculated as:

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Benefit of using dB

- One reason that engineers use the decibel to measure the changes in signal strength is that decibel numbers can be added (or subtracted) when we are measuring several points instead of just two.



dBm

- dBm is used to measure signal power in milliwatts
- $\text{dBm} = 10 \log_{10} P_m$
- P_m : power in milliwatt unit

- $1\text{mW} = 0\text{dBm}$
- $10\text{mW} = 10\text{dBm}$
- $100\text{mW} = 20\text{dBm}$
- $0.1\text{mW} = -10\text{dBm}$

Example

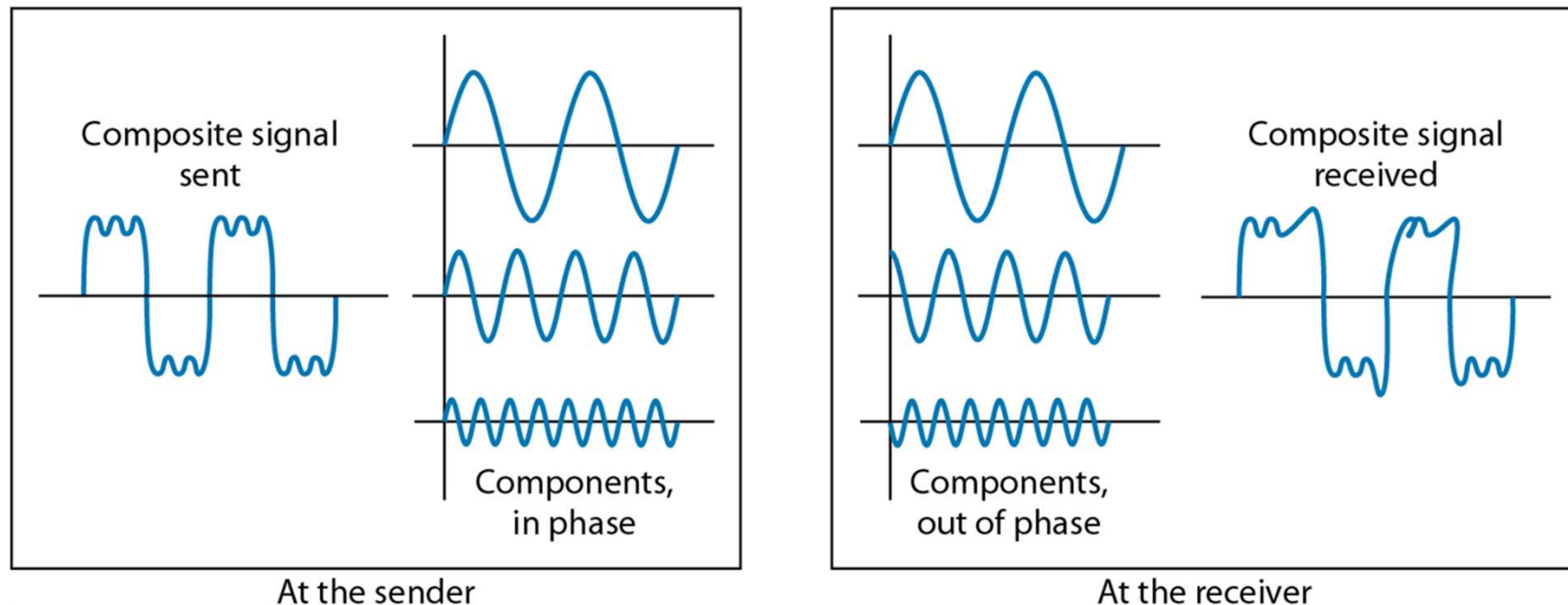
- The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3dB/km has a power of 2mW, what is the power of signals at 5km?

$$\text{dB} = 10 \log_{10} (P_2 / P_1) = -1.5 \quad \longrightarrow \quad (P_2 / P_1) = 10^{-0.15} = 0.71$$

$$P_2 = 0.71P_1 = 0.7 \times 2 \text{ mW} = 1.4 \text{ mW}$$

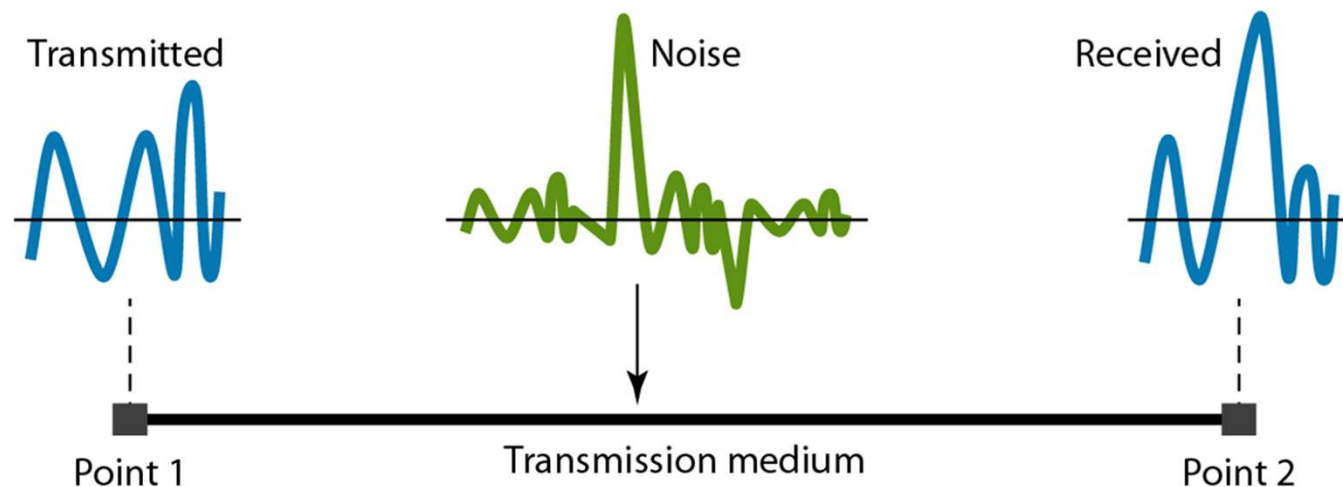
Distortion

- Different frequency components of a signal travel at different speeds through a medium
- This causes them to arrive at the receiver with different delays (phase shifts)
- The recombined signal at the receiver has a different shape from the original
- Can be compensated using an equalizer at the receiver



Noise

- Unwanted signals that get added to the transmitted signal
- Unlike distortion, noise is random and unpredictable - cannot be completely eliminated
- Types of noise
 - Thermal noise: random motion of electrons in a conductor
 - Induced noise: from external sources such as motors, power lines
 - Crosstalk: interference from adjacent wires or channels
 - Impulse noise: sudden spikes (e.g., lightning, power surges)
- Noise is the ultimate factor that limits communication performance



Signal-to-Noise Ratio (SNR)

- $SNR = \text{signal power} / \text{noise power}$
- $SNR_{dB} = 10 \log_{10}(SNR)$
- High SNR $\rightarrow$ signal is much stronger than noise $\rightarrow$ better quality
- Low SNR $\rightarrow$ noise is comparable to signal $\rightarrow$ errors increase
- SNR determines the maximum achievable data rate (Shannon theorem)

Exercise

- The power of a signal is 10mW and the power of the noise is 1uW; what are the values of SNR and SNR_{dB} ?

Example

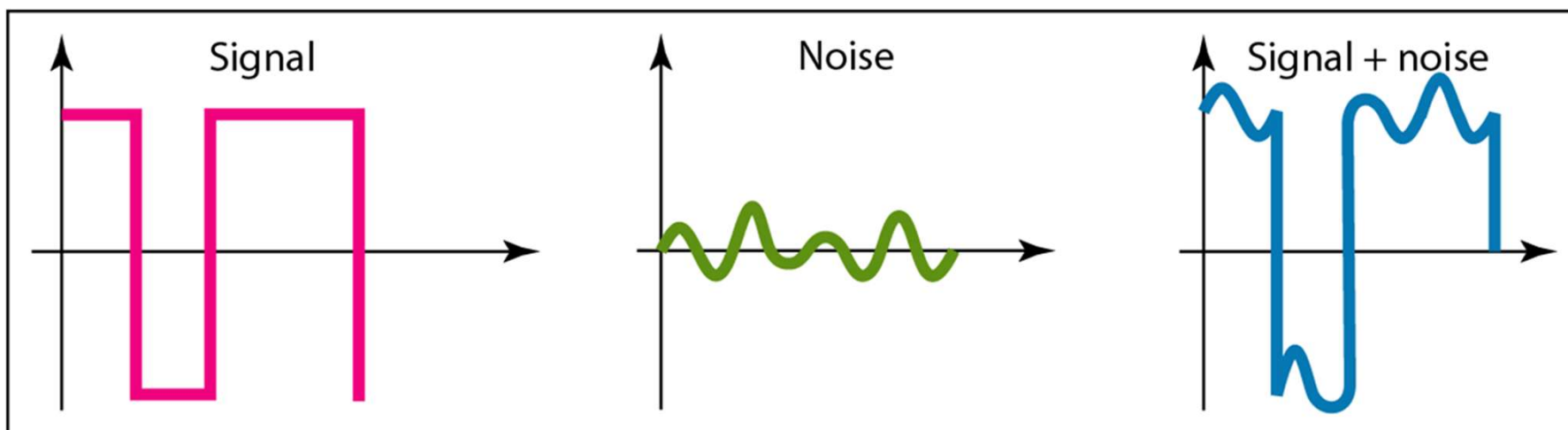
- What is the SNR and SNR_{dB} for a noiseless channel?

$$\text{SNR} = (\text{signal power}) / 0 = \infty \longrightarrow \text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

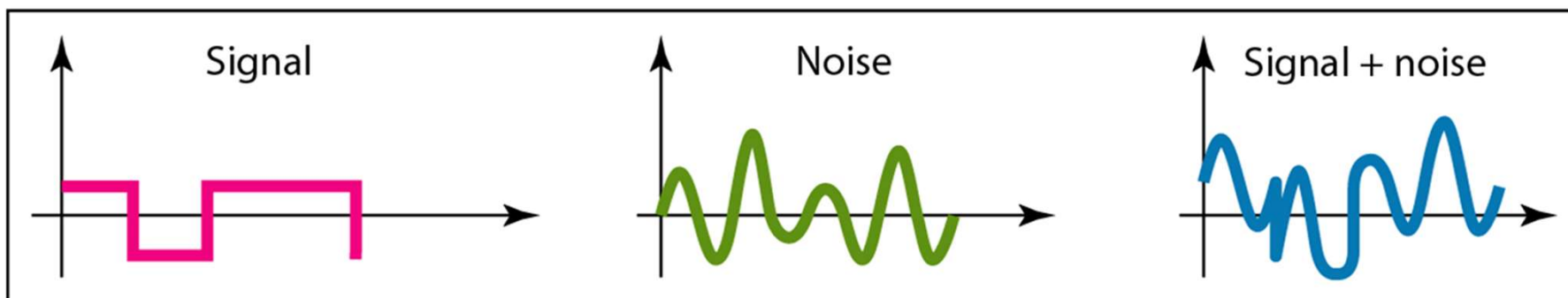
- This is an ideal case; it can never happen in real life.

Impact of noise on signal

- If SNR is too low, signal cannot be recovered



a. Large SNR



b. Small SNR

Data Rate Limits

Data Transfer Rate (Bit Rate)

- Data transfer rate is dependent upon the following three:
 - Bandwidth
 - Number of signal levels
 - Channel quality (noise level)
- Nyquist theorem addresses the first two (noiseless channel)
- Shannon theorem addresses all three (noisy channel)

Nyquist Equation

- Assumes noiseless channel
- Data rate = $2 \times B \times \log_2 L$
 - B: bandwidth, L: number of signal levels
- When L=2: Data rate = 2B
 - minimum bandwidth = N/2
- What if we use too many signal levels?
 - Levels become closer together → more vulnerable to noise
 - This is why we also need the Shannon theorem (noisy channel)

Exercise

- Consider a noiseless channel with a bandwidth of 3000Hz transmitting a signal with two signal levels. What is the maximum bit rate?
- Consider a noiseless channel with a bandwidth of 3000Hz transmitting a signal with four signal levels. What is the maximum bit rate?

Example

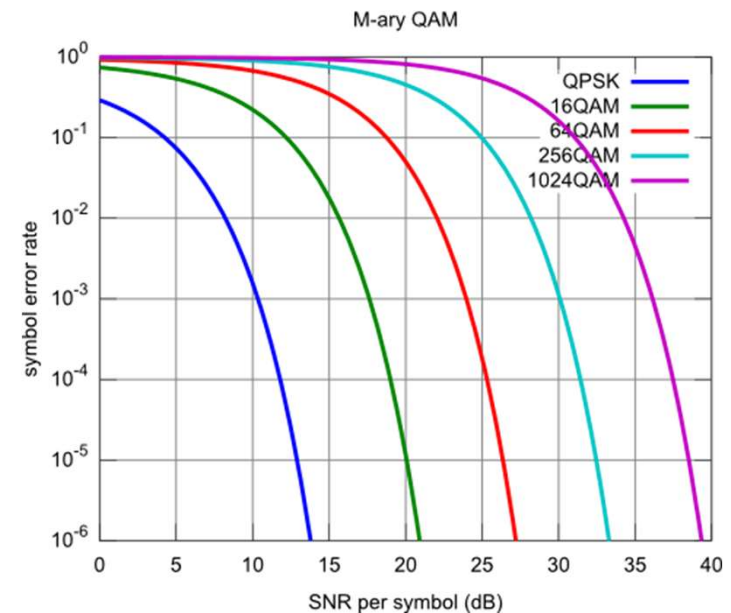
- We need to send 265kbps over a noiseless channel with a bandwidth of 20kHz. How many signal levels do we need?

$$265,000 = 2 \times 20,000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

- In practice, L is chosen as a power of 2 (so each level maps to an integer number of bits)
- $L = 128$ levels (7 bits per level)
- Actual data rate: $2 \times 20,000 \times 7 = 280\text{kbps}$

Shannon Capacity

- Maximum bit rate in a noisy channel
- $C = B \times \log_2(1+\text{SNR})$
 - C: capacity (maximum bit rate)
 - B: bandwidth
 - SNR: signal-to-noise ratio (linear, not in dB)



- Nyquist says: increase levels → increase data rate (no limit)
- Shannon says: noise sets an absolute upper limit on data rate, regardless of the number of levels
- Graph: higher-order modulation (more levels) achieves higher data rate, but requires higher SNR to maintain the same error rate

Example

- Consider an extremely noisy channel in which SNR is zero. What is the maximum bit rate for this channel?

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0) = B \log_2 1 = B \times 0 = 0$$

- In other words, we cannot send or receive any data through this channel.

Example

- We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000Hz (300 – 3300Hz) assigned for data communications. The signal-to-noise ratio is usually 3162.
- The channel capacity is:

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34,860 \text{ bps}$$

Example

- Suppose SNR is 36dB, and channel bandwidth is 2MHz. What is the channel capacity?

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \longrightarrow \text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \longrightarrow \text{SNR} = 10^{3.6} = 3981$$

$$C = B \log_2(1 + \text{SNR}) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Example

- Approximation for High SNR
 - When SNR is high, $SNR + 1 \approx SNR$, so $C = B \times \log_2 SNR$
 - Since $\log_2 SNR = \log_2(10) \times \log_{10}(SNR)$
 - $= 3.32 \times \left(\frac{SNR_{dB}}{10}\right) \approx \frac{SNR_{dB}}{3}$
- Simplified formula: $C = B \times (SNR_{dB} / 3)$
- Quick estimation: for the previous example,
 - $C = 2MHz \times (36 / 3) = 24Mbps$

Performance

Bandwidth

- Two meanings of bandwidth
 - Frequency bandwidth: the range of frequencies a channel can pass (Hz)
 - Data bandwidth: the maximum bit rate a channel can support (bps)
- These are related: a wider frequency bandwidth generally shows a higher data rate (as shown by Nyquist and Shannon)
- Example
 - Bandwidth of a line is 4kHz. This line has 56Kbps of bandwidth.

Throughput

- The actual data transfer rate (bps)
 - Throughput < Bandwidth
 - due to overhead (headers, retransmissions), congestion, protocol inefficiency, etc.
- Example
 - A network with bandwidth of 10Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

$$\text{Throughput} = (12,000 \times 10,000) / 60 = 2 \text{ Mbps}$$

- Throughput of this network is 1/5 of its bandwidth

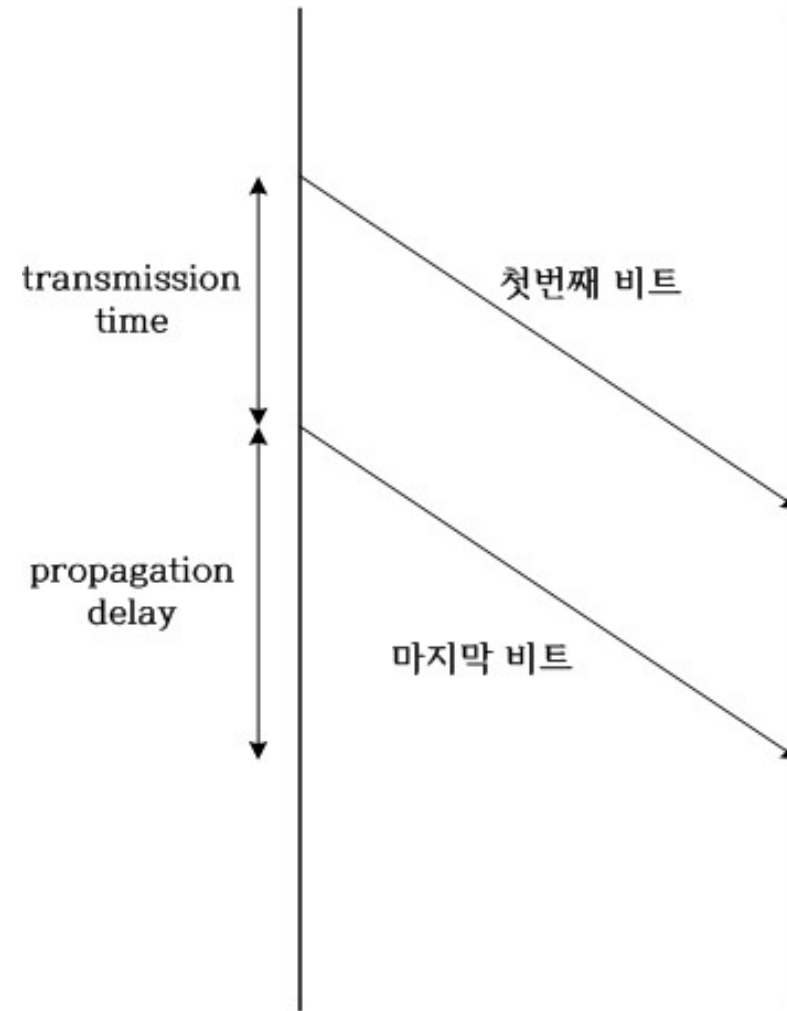
Delay

- Duration between the time the first bit of a message leaves the source and the time the last bit of the message is received at the destination
- Delay = transmission delay + propagation delay + queuing delay + processing delay

Delay Components

- Transmission delay
 - Time to push all bits of a message onto the link
 - $\text{message size} / \text{bandwidth (data rate)}$
 - Depends on message size and link speed
- Propagation delay
 - Time for a bit to travel from source to destination
 - $\text{distance} / \text{propagation speed}$
 - Depends on distance and medium
- Queuing delay
 - Time waiting for other packets ahead in the queue (at routers/switches)
- Processing delay
 - Time for a system to process the message (e.g., error checking, routing decisions)

Delay Components



Example

- What is the propagation time if the distance between the two points is 12,000km? Assume the propagation speed to be $2.4 \times 10^8 \text{m/s}$ in cable.

$$\text{Propagation time} = (12,000 \times 1000) / (2.4 \times 10^8) = 50\text{ms}$$

- A bit can go over the Atlantic Ocean in only 50ms if there is a direct cable

Example

- What are the propagation time and the transmission time for a 2.5KB message if the bandwidth of the network is 1Gbps? Assume that the distance between the sender and the receiver is 12,000km and that the signal travels at 2.4×10^8 m/s

$$\text{Propagation time} = (12,000 \times 1000) / (2.4 \times 10^8) = 50 \text{ ms}$$

$$\text{Transmission time} = (2500 \times 8) / 10^9 = 0.020 \text{ ms}$$

- Message is short and the bandwidth is large
→ Propagation time is the dominant factor

Example

- What are the propagation time and the transmission time for a 5MB message if the bandwidth of the network is 1Mbps? Assume that the distance between the sender and the receiver is 12,000km and the signal travels at 2.4×10^8 m/s

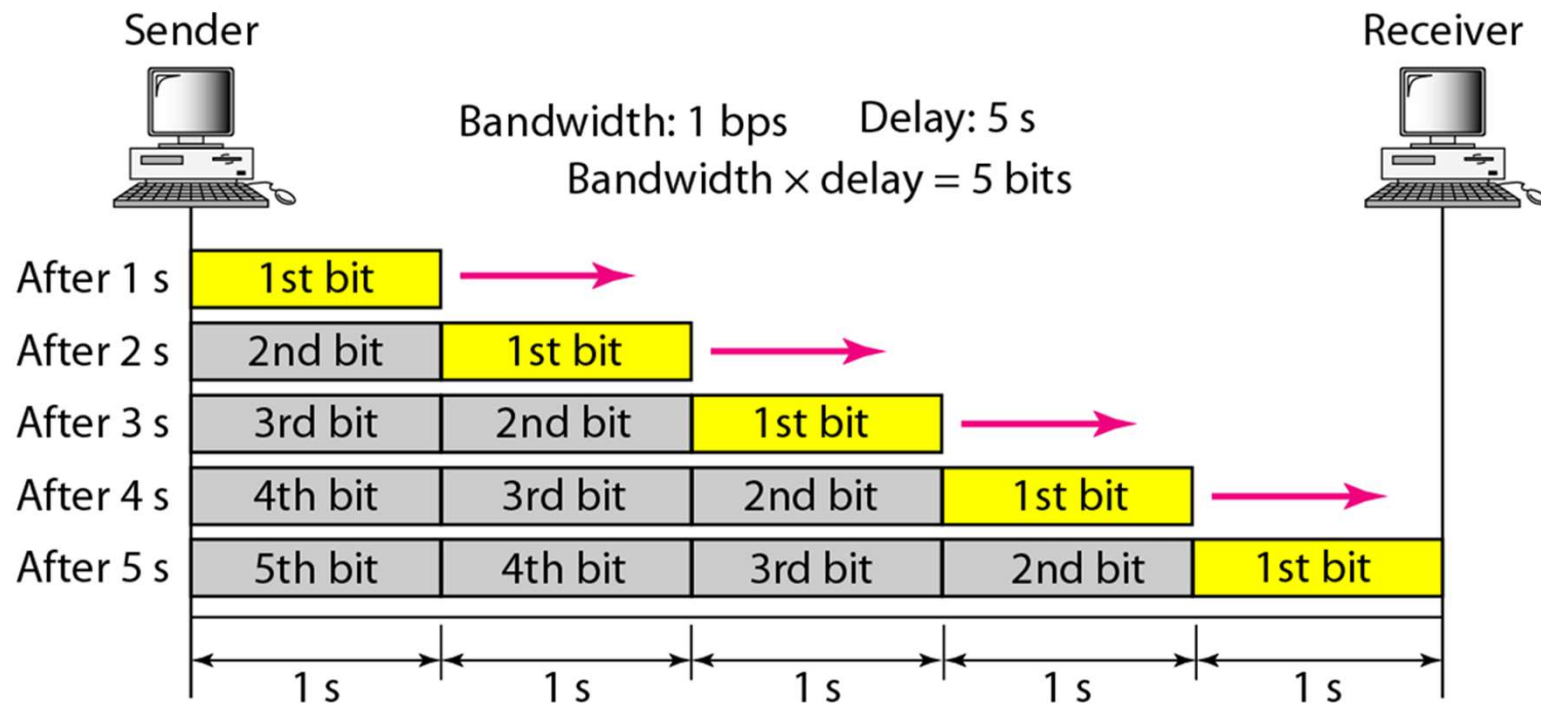
$$\text{Propagation time} = (12,000 \times 1000) / (2.4 \times 10^8) = 50 \text{ ms}$$

$$\text{Transmission time} = (5,000,000 \times 8) / 10^6 = 40 \text{ s}$$

- Message is long and bandwidth is small
→ Transmission time is the dominant factor

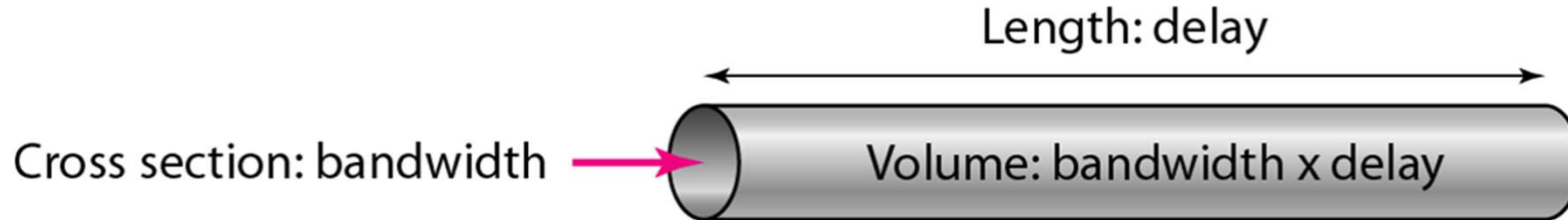
Bandwidth-Delay Product

- Bandwidth-delay product represents the "volume" of the channel
 - How much data can be fit into the channel between sender and receiver
 - This concept is important for efficient data transfer: the sender should keep this "pipe" full to maximize throughput



Bandwidth-Delay Product

- The channel can be thought as a pipe
- Bandwidth-delay product is the number of bits that fills up the pipe



Exercise

- Suppose bandwidth of a link is 10Mbps, and the delay is 50ms. What is the amount of data to fill up this channel?

End of Chapter
